Estimating beta-function selection models in meta-analysis with dependent effects

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Highlights

What is already known

What is new

Potential impact for RSM readers

Estimating beta-function selection models in meta-analysis with dependent effects

Meta-analysis is a critical tool for synthesizing evidence across studies to draw generalizable conclusions in fields such as education, psychology, and medicine. However, the validity of meta-analytic findings depends on the completeness and representativeness of the available data from primary studies. A persistent threat to validity is selective reporting, where effect size estimates are more likely to be published, and thus included in a meta-analysis, if they are statistically significant and consistent with researchers' hypotheses (Carter, Schönbrodt, Gervais, & Hilgard, 2019). Extensive evidence from diverse domains suggests that such reporting biases are widespread (e.g., O'Boyle Jr, Banks, & Gonzalez-Mulé, 2017; Chan, Hróbjartsson, Haahr, Gøtzsche, & Altman, 2004; Franco, Malhotra, & Simonovits, 2016; John, Loewenstein, & Prelec, 2012; Lancee, Lemmens, Kahn, Vinkers, & Luykx, 2017; Pigott, Valentine, Polanin, Williams, & Canada, 2013), potentially distorting estimates of intervention effects and leading to unwarranted conclusions.

To address selective reporting bias, a variety of statistical approaches have been developed (Marks-Anglin & Chen, 2020; Rothstein, Sutton, & Borenstein, 2005). Among the most promising are p-value selection models, which assume that the probability of an effect being reported depends on its statistical significance. These models are appealing because they are based on explicit assumptions about the selection mechanism and can be integrated into conventional meta-analytic frameworks (e.g., meta-regression). The step-function selection model, originally developed by Hedges (1992) and later extended by Vevea and Hedges (1995), assumes a piece-wise constant selection probability across different p-value thresholds (e.g., $\alpha = .05$). This approach captures plausible patterns of selective reporting and has been shown to outperform simpler diagnostics or regression-based adjustments, particularly when effect sizes are heterogeneous (Carter et al., 2019; Terrin, Schmid, Lau, & Olkin, 2003).

Most existing methods for assessing and/or correcting for selective reporting—including *p*-value selection models—have been developed under the assumption that each study contributes a single, independent effect size. However, this assumption is implausible in many contemporary research syntheses. Many meta-analyses now include multiple, dependent effect sizes per study, such as estimates derived from different outcomes, time points, or treatment comparisons. Recent work addressed this gap by integrating the step-function selection model with robust variance estimation (RVE) or bootstrapping techniques to account for dependent effect sizes (Pustejovsky, Citkowicz, & Joshi, 2025). Following the logic of generalized estimating equations with working independence structures for the analysis of longitudinal data (Liang & Zeger, 1986), this approach focuses on modeling the marginal distribution of the effect size estimates. Each effect size estimate is treated as if it were an independent observation for the purposes of estimation, with RVE or bootstrapping used to account for unmodeled dependence. Simulation studies demonstrated that this approach reduces bias in the estimate of the overall effect size, but that there is a bias-variance trade-off relative to the unadjusted meta-analytic model. Moreover, simulations showed that a two-stage cluster bootstrapping technique can be used to generate confidence intervals with coverage rates that approach the nominal level of 0.95 in meta-analyses with a moderate number of studies.

While the step-function selection model provides a structured and intuitive way to characterize selective reporting, it relies on the meta-analyst to specify "psychologically salient" *p*-value thresholds, such as 0.05 or 0.01. In practice, the true pattern of selection may not conform neatly to such step-wise forms. To address this limitation, we introduce a selection model with a truncated beta density function to model the selection process, allowing the selection probability to vary smoothly as a function of the *p*-value of each effect size estimate. This model builds on earlier work by Citkowicz and Vevea (2017), extending it to the context of dependent effect sizes by incorporating RVE and bootstrap methods. The beta-density selection model offers a more flexible alternative for capturing diverse forms of selection without requiring the meta-analyst to identify specific thresholds a priori. Furthermore, this approach allows meta-analysts to assess whether the form of selection matters—a critical extension for determining the robustness of findings to different specifications of the selection function.

We begin the paper by sketching the beta-density selection model and describing estimation and inference procedures. We then illustrate the model using a previously published meta-analysis, highlighting how it can reveal patterns of selective reporting not captured by the step-function selection model. Next, we report findings from an extensive simulation study that evaluates the model's performance under a variety of conditions and that investigates whether the form of selection matters. We conclude with summary findings, limitations, directions for future research, and implications for practice.

Models and Estimation Methods

Selection models are comprised of two components. The first component, the *evidence-generating process*, describes the distribution of effect sizes before selection, typically using a conventional random-effects model or meta-regression model. The second component, the *selection process*, identifies how the distribution is changed based on the likelihood of an effect size being reported. The combined model includes parameters that define the selection process, along with parameters capturing the mean and heterogeneity of the effect size distribution that are adjusted for selective reporting.

Following the approach outlined in previous work on step-function models (Pustejovsky, Citkowicz, et al., 2025), we model the marginal distribution of effect size estimates rather than the joint distribution within studies. In other words, the model treats each effect size as independent, ignoring its relationship with other effect sizes from the same study. We adopt this marginal modeling strategy for several reasons. First, it provides a parsimonious and plausible framework for selection based on the significance of individual effect sizes—often considered a primary driver of selective reporting bias—without requiring unknown assumptions about joint selection probabilities across multiple effects. Second, focusing on the marginal distribution eases computational tractability by avoiding the high-dimensional integration required for multivariate selection models. Third, it has strong precedents in other areas of statistical analysis, including generalized estimating equations with working independence structures for longitudinal data (Liang & Zeger, 1986) and pseudo-likelihood methods for multivariate meta-analysis (Y. Chen, Hong, & Riley, 2015). To account for dependence among effect sizes, we pair this marginal model with cluster-robust variance estimation (RVE) or clustered bootstrapping techniques. Although this approach limits interpretation to the marginal distribution and does not distinguish between study-level and outcome-level selection, it remains a practical and plausible framework for modeling selective reporting based on the statistical significance of individual estimates.

We use the following notation to describe the model and estimation procedures. Consider a meta-analytic dataset comprising J studies, where study j reports k_j effect size estimates. Let y_{ij} denote the ith effect size estimate from study j, with associated standard error σ_{ij} and one-sided p-value p_{ij}. The one-sided p-value is defined relative to the null hypothesis that the true effect size is less than or equal to zero, with alternative hypothesis that the effect size is positive. Let $\mathbf{x}_{ij}$ be a $1 \times x$ row vector of predictors representing characteristics of the effect size, sample, or study procedures. We use $\Phi()$ to denote the standard normal cumulative distribution function and $\phi()$ to denote the standard normal density function.

Evidence-generating process

We assume an evidence-generating process based on a standard random-effects meta-regression model. Let Y^* denote a potentially reported effect size estimate, with standard error σ^*, one-sided p-value p^*, and predictor vector $\mathbf{x}^*$. Then the evidence-generating process is defined as

$$(Y^*|\sigma^*, \mathbf{x}^*) \sim N\left(\mathbf{x}^*\boldsymbol{\beta},\ \tau^2 + \sigma^{*2}\right), \tag{1}$$

where $\boldsymbol{\beta}$ is an $x \times 1$ vector of regression coefficients and τ^2 is the marginal variance of the effect size distribution. This model treats effect sizes as independent and characterizes *total* heterogeneity without decomposing within- and between-study variation.

Selection process

A p-value selection process is defined by a selection function that specifies the probability that an effect size is reported, conditional on its p-value. Letting O indicate whether Y^* is observed, a selection function involves specifying $\Pr(O = 1|p^*) \propto w(p^*; \boldsymbol{\lambda})$ for some strictly positive function $w(.; \boldsymbol{\lambda})$ on the interval $[0, 1]$ with an unknown parameter vector $\boldsymbol{\lambda}$.

Citkowicz and Vevea Citkowicz and Vevea (2017) proposed a selection function based on a truncated beta density with two parameters, providing flexibility to capture diverse selection patterns more parsimoniously than with a step function model. Because the beta density can be unbounded near 0 and 1, they proposed truncating it to make the model computationally tractable, assuming constant selection probabilities for p-values in the range $[0, \alpha_1]$ and $[\alpha_2, 1]$. Given these pre-specified thresholds α_1 and α_2 and selection parameters $\boldsymbol{\lambda} = (\lambda_1, \lambda_2)$, the beta density selection function is expressed by

$$w(p^*, \boldsymbol{\lambda}) = \begin{cases} \alpha_1^{\lambda_1 - 1}(1-\alpha_1)^{\lambda_2 - 1} & \text{if} \quad p^* \leq \alpha_1 \\ (p^*)^{\lambda_1 - 1}(1-p^*)^{\lambda_2 - 1} & \text{if} \quad \alpha_1 < p^* < \alpha_2 \\ \alpha_2^{\lambda_1 - 1}(1-\alpha_2)^{\lambda_2 - 1} & \text{if} \quad \alpha_2 \leq p^*. \end{cases} \tag{2}$$

Equation (2) can be written equivalently as

$$w(Y_i^*, \sigma_i^*, \boldsymbol{\lambda}) = \begin{cases} \alpha_1^{\lambda_1 - 1}(1-\alpha_1)^{\lambda_2 - 1} & \text{if} \quad \sigma^*\Phi^{-1}(1-\alpha_1) \leq Y^* \\ \left[\Phi\left(-Y^*/\sigma^*\right)\right]^{\lambda_1 - 1}\left[\Phi\left(Y^*/\sigma^*\right)\right]^{\lambda_2 - 1} & \text{if} \quad \sigma^*\Phi^{-1}(1-\alpha_2) < Y^* < \sigma^*\Phi^{-1}(1-\alpha_1) \\ \alpha_2^{\lambda_1 - 1}(1-\alpha_2)^{\lambda_2 - 1} & \text{if} \quad Y^* \leq \sigma^*\Phi^{-1}(1-\alpha_2). \end{cases} \tag{3}$$

(a) *Decreasing selection:* $\lambda_1 = 0.5, \lambda_2 = 2.0$ (b) *Complex selection:* $\lambda_1 = 0.5, \lambda_2 = 0.6$

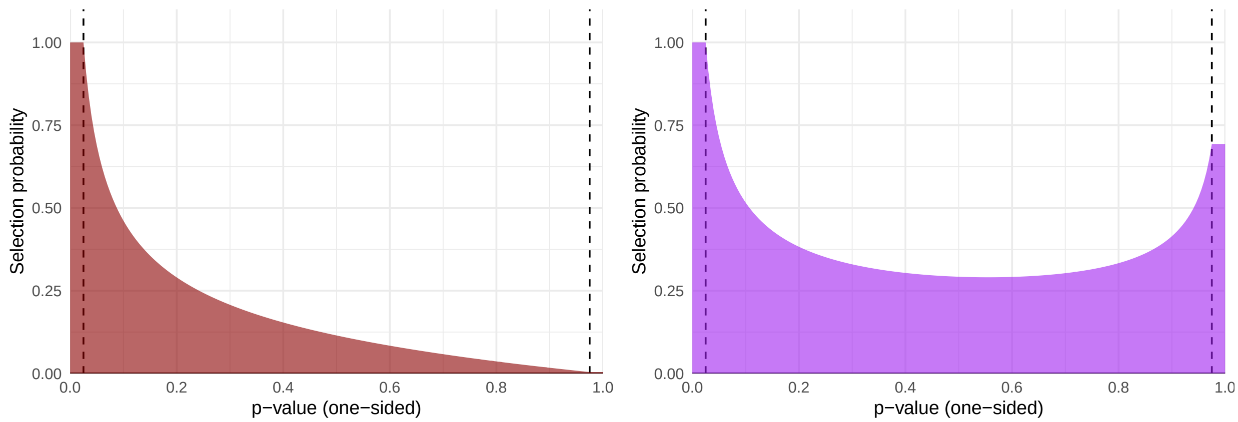

Figure 1
Examples of beta density functions

When $\lambda_1 = \lambda_2 = 1$, the selection function is flat and selective reporting is absent.

Citkowicz and Vevea (Citkowicz & Vevea, 2017) used extreme truncation points ($\alpha_1 = 10^{-5}$, $\alpha_2 = 1 - 10^{-5}$), but such choices can make the model overly sensitive to rare, extreme p-values, potentially producing implausible estimates (Hedges, 2017). We instead consider more moderate, psychologically salient thresholds of $\alpha_1 = .025$ and $\alpha_2 = .975$. These thresholds correspond to the standard two-sided $\alpha = .05$ level and help stabilize estimation by preventing the selection function from being determined by a few extreme p-values, while still allowing the model to capture diverse selection patterns.

Figure 1 depicts several shapes that the beta density can assume. Figure 1a. shows a curve with $\lambda_1 = 0.5$ and $\lambda_2 = 2.0$, corresponding to a preference for statistically significant findings and a strictly decreasing probability of selection. Figure 1b. depicts a curve with $\lambda_1 = 0.5$ and $\lambda_2 = 0.6$, leading to a more complex form of selection in which both significantly positive and significantly negative effect sizes are more likely to be reported than null effects.

Distribution of observed effect size estimates

The combined model for the marginal density of an observed effect size estimate y_{ij} with standard error σ_{ij} has the form

$$f(Y = y_{ij}|\sigma_{ij}, \mathbf{x}_{ij}) = \frac{w\left(y_{ij}, \sigma_{ij}; \boldsymbol{\lambda}\right)}{A(\mathbf{x}_{ij}, \sigma_{ij}; \boldsymbol{\beta}, \tau^2, \boldsymbol{\lambda})} \times \frac{1}{\sqrt{\tau^2 + \sigma_{ij}^2}} \phi\left(\frac{y_{ij} - \mathbf{x}_{ij}\boldsymbol{\beta}}{\sqrt{\tau^2 + \sigma_{ij}^2}}\right), \tag{4}$$

where

$$A(\mathbf{x}_{ij}, \sigma_{ij}; \boldsymbol{\beta}, \tau^2, \boldsymbol{\lambda}) = \int_{\mathbb{R}} w\left(y, \sigma_{ij}; \boldsymbol{\lambda}\right) \times \frac{1}{\sqrt{\tau^2 + \sigma_{ij}^2}} \phi\left(\frac{y - \mathbf{x}_{ij}\boldsymbol{\beta}}{\sqrt{\tau^2 + \sigma_{ij}^2}}\right) dy. \tag{5}$$

The $A(\mathbf{x}_{ij}, \sigma_{ij}; \boldsymbol{\beta}, \tau^2, \boldsymbol{\lambda})$ term in (4) must be computed using numerical integration. If $w(y, \sigma; \boldsymbol{\lambda}) = 1$, then $A(\mathbf{x}, \sigma; \boldsymbol{\beta}, \tau^2, \boldsymbol{\lambda}) = 1$ and there is no selective reporting. The density then reduces to the unweighted density of the evidence-generating process and the $\boldsymbol{\beta}$ estimates from the adjusted beta function selection model will approximate those of a standard random effects model.

Estimation Method

We estimate model parameters using the composite marginal likelihood, with an additional penalty term for regularization that functions similarly to a prior in Bayesian inference; thus, we call this approach penalized maximum likelihood (PML). The PML estimator treats each observed effect size estimate as if it were mutually independent for the purposes of estimation, following the strategy validated with step function selection models (Pustejovsky, Citkowicz, et al., 2025) and consistent with established composite likelihood approaches (Cox & Reid, 2004; Lindsay, 1988; Varin, 2008). To address potential numerical instability and ensure the plausibility of results—particularly in meta-analyses with a small number of studies or sparse data near the p-value thresholds—we incorporate weak penalties into the composite likelihood. Specifically, we apply a log-F(2, 2) penalty to the heterogeneity parameter τ^2 and independent normal penalties on the log-transformed selection parameters, $\log(\lambda_1) \sim N(0,1)$ and $\log(\lambda_2) \sim N(0,1)$. These penalties provide a form of stabilization that prevents estimates from reaching extreme or unstable values during the optimization process without strongly influencing the results when the data are informative.

Estimation proceeds by maximizing the penalized log-composite-likelihood function. To account for the statistical dependence between effect sizes within the same study, we construct confidence intervals using robust (sandwich-type) variance estimators based on study-level score contributions. The sandwich estimator used for the beta-density selection model follows the same structural form as the one developed for the step-function selection model. Online Appendix A provides additional technical details about computing the PML estimator and sandwich estimator.

Bootstrap inference

To improve inference accuracy with a limited number of studies, we also implement bootstrapping procedures, which generate pseudo-samples through random resampling or reweighting of the original data. In prior work (Pustejovsky, Citkowicz, et al., 2025), we considered several bootstrap methods designed to preserve the dependence structure across clusters, including the non-parametric clustered bootstrap, two-stage bootstrap (Field & Welsh, 2007; Leeden, Meijer, & Busing, 2008), and fractional random weight bootstrap (Rubin, 1981; Xu, Gotwalt, Hong, King, & Meeker, 2020). Finding from this prior work indicated that the two-stage bootstrap consistently outperformed the alternatives, and so here we focus exclusively on two-stage bootstrapping. Similarly, after testing several standard bootstrap-based methods for computing confidence intervals, including percentile, basic, studentized, and bias-corrected-and-accelerated (Davison & Hinkley, 1997; Efron, 1987), we chose the percentile bootstrap method as it consistently yielded the best coverage rates. These resampling-based procedures are particularly useful in small-sample contexts where sandwich estimators may perform poorly.

Empirical Example

To demonstrate the interpretation of the beta-function selection model in practice, we reanalyzed the data from a meta-analysis of leader group prototypicality (the degree to which a leader is perceived to embody a group's shared identity), conducted by Steffens and colleagues (Steffens, Munt, Knippenberg, Platow, & Haslam, 2021). The meta-analytic sample included findings from 128 studies with 251 effects. Approximately half of the studies contributed multiple effects per study, due to either multiple samples or multiple outcomes, leading to dependent effect sizes. Effect sizes were measured as correlations (converted to Fisher's z-scores for analysis), with positive effects indicating that leaders who are seen as more prototypical of their group tend to be more effective.

We conducted the analyses using R Version 4.5.3 (R Core Team, 2023). We first used the `rma.uni()` function from the `metafor` package to fit a standard random-effects model based on an independent effects working model with random effects for each effect size (Viechtbauer, 2010), with standard errors clustered by study. The overall estimate of the average effect using the unadjusted model was 0.42, cluster-robust 95% CI [0.36, 0.47], significantly different from zero ($p < 0.001$). The unadjusted estimate of heterogeneity was $\tau = 0.29$.

To adjust for selective reporting, we used the `selection_model()` function from the `metaselection` package (Pustejovsky, Joshi, & Citkowicz, 2025) to fit the beta-density selection model (with default truncation thresholds of $\alpha_1 = .025, \alpha_2 = .975$) and two different step-function selection models—a one-step model ($\alpha_1 = 0.025$) and a two-step model ($\alpha_1 = .025, \alpha_2 = .5$). Consistent with the methods described above, all three selection models were estimated using the PML approach and an independent-effects working model, with two-stage cluster bootstrapping ($B = 1999$ bootstraps) used to account for dependent effect sizes.

Table 1

Selection model parameter estimates fit to meta-analysis of leader group prototypicality from Steffens et al. (2021)

	Beta density		One-step		Two-step	
Parameter	Est. (SE)	95% CI	Est. (SE)	95% CI	Est. (SE)	95% CI
Summary meta-analysis						
β_0	0.28 (0.08)	[0.07, 0.43]	0.42 (0.03)	[0.34, 0.49]	0.37 (0.05)	[0.25, 0.46]
τ	0.38 (0.18)	[0.30, 0.46]	0.29 (0.12)	[0.25, 0.33]	0.33 (0.13)	[0.28, 0.38]
λ_1	0.99 (0.14)	[0.69, 1.32]	1.08 (0.30)	[0.53, 2.13]	1.09 (0.25)	[0.60, 1.93]
λ_2	1.73 (0.14)	[1.25, 2.52]			0.36 (0.51)	[0.11, 1.06]
Moderator analysis						
β_0	0.01 (0.12)	[-0.31, 0.24]	0.26 (0.05)	[0.16, 0.37]	0.16 (0.07)	[0.01, 0.30]
β_1	0.33 (0.10)	[0.14, 0.58]	0.19 (0.05)	[0.08, 0.29]	0.25 (0.07)	[0.10, 0.39]
τ	0.36 (0.17)	[0.29, 0.44]	0.28 (0.12)	[0.24, 0.32]	0.32 (0.14)	[0.27, 0.36]
λ_1	0.97 (0.14)	[0.67, 1.30]	1.04 (0.30)	[0.51, 2.05]	1.06 (0.25)	[0.60, 1.86]
λ_2	1.74 (0.15)	[1.24, 2.56]			0.34 (0.51)	[0.11, 0.96]

Note:
Est. = estimate; SE = bootstrap standard error; CI = percentile boodstrap confidence interval.

The upper panel of Table 1 reports parameter estimates from all three models, along with bootstrap standard errors and 95% confidence intervals. Across the three selection models, the estimated average effect sizes are all positive and significant, but they vary in magnitude and interpretation. Compared to the unadjusted model, the average adjusted effect is 33% smaller when using the beta-density selection model ($ES = 0.28$). In contrast, the adjustment is minimal for the one-step selection model, with an average effect estimate that is 2% larger than under the unadjusted model ($ES = 0.42$); the two-step selection model yields an estimate that is 12% smaller ($ES = 0.37$) than the unadjusted average effect.

These differences are driven by the specific form of the selection curve estimated from each model, as depicted in Figure 2. The beta-density model identifies a substantial downward adjustment, capturing a smooth, continuous decline in the likelihood of reporting as effects move toward non-significance. The one-step model, on the other hand, estimates a nearly flat selection process, suggesting that when only one p-value threshold is considered,

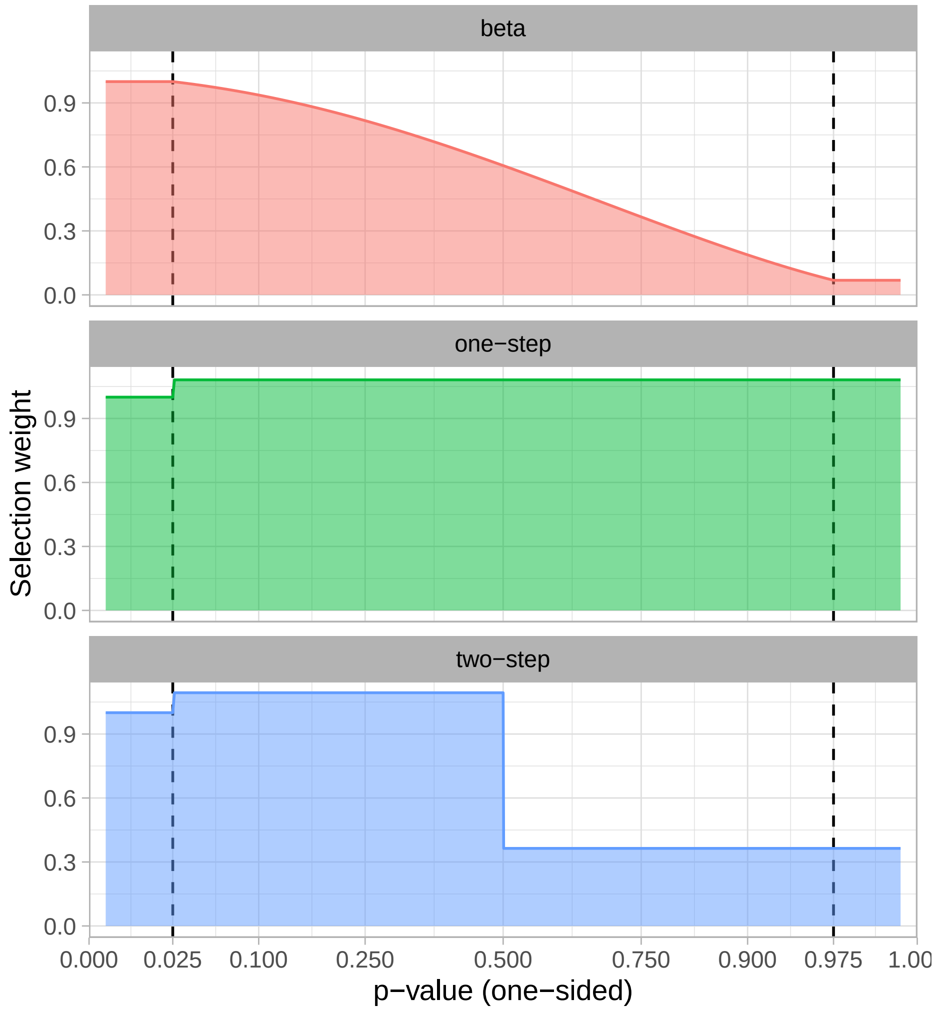

Figure 2

Estimated selection functions based on beta density, one-step, and two-step selection models

the overall evidence for selective reporting appears negligible. The two-step model identifies a slight increase in reporting probability at the $\alpha_1 = .025$ threshold, followed by a substantial decrease at the $\alpha_2 = .05$ threshold. Notably, the beta-density model provided the best fit to the data (Log composite likelihood = -49.02), followed by the two-step model (-52.94) and the one-step model (-58.12). This divergence illustrates how the continuous functional form of the beta-density model can capture nuanced reporting patterns—such as a smooth decline in the likelihood of reporting for increasingly negative results—that may be obscured or dependent on the pre-defined thresholds of a step-function model.

A key advantage of selection models over other methods for diagnosing and adjusting for selective reporting bias is that they allow for inclusion of potential moderator variables as predictors of average effect size. This feature allows meta-analysts to investigate whether selective reporting bias varies across study characteristics or, conversely, whether suspected moderator effects persist after adjusting for selection. In the original meta-analysis, Steffens and colleagues (Steffens et al., 2021) examined whether the relationship between prototypicality and leadership outcomes was moderated by the target leader's role. Specifically, they coded whether the target person occupied a formal leadership role (e.g., a supervisor, political leader, or CEO) or held a nonformal leadership position (e.g., a group member or candidate for a role). They found that the leader prototypicality–outcome relationship is significantly stronger for leaders in formal leadership roles than for those occupying nonformal roles. We investigated whether this moderating effect remains after adjusting for selective reporting.

We reran all four models using target leader's role as a moderator (Steffens et al., 2021). Based on a random effects meta-regression without adjustment for selective reporting, the relationship for leaders in nonformal roles produced an average effect size of 0.25, 95% CI [0.16, 0.35]. The average effect for leaders in formal roles was larger by 0.19 SD, 95% CI [0.08, 0.30], which was statistically distinct from zero ($p = 0.0014$). The unadjusted estimate of heterogeneity was $\tau = 0.28$, similar to the summary meta-analysis.

The lower panel of Table 1 reports parameter estimates from the corresponding selection models that include target leader's role as a predictor. The β_1 coefficient estimates represent differences in average effects between leaders in formal versus nonformal roles, which were all statistically significant. Compared to the unadjusted model's estimated difference of 0.19, the beta-density model estimated a substantially larger difference of 0.33, whereas the two-step model yielded a difference of 0.25 and the one-step model remained unchanged at 0.19. This suggests that the moderating effect of leader status is robust to selective reporting; however, the advantage for formal leaders may actually be underestimated when selective reporting bias is ignored. Moreover, the selection models lead to different conclusions about the average effects for leaders in nonformal roles (β_0). The estimate from the beta-density model is not statistically distinct from zero and is substantially smaller than the estimate from the unadjusted model. In contrast, the estimated average effects for the reference group in the step-function selection models remain statistically significant, though they vary in magnitude. Selection parameter estimates are very similar to the estimates from the corresponding summary meta-analysis models.

The results of this application demonstrate that the choice of a smooth versus discrete selection function can lead to different conclusions regarding the magnitude and significance of adjusted effects. The beta-density selection model's ability to capture a continuous range of selection probabilities provides a flexible alternative to the step-wise form of selection assumed by step-function selection model, particularly when reporting bias does not follow rigid significance thresholds. The fact that the beta-density model identified a larger moderator effect and a near-zero baseline effect highlights how a continuous function can reveal nuanced patterns that may be obscured by the pre-defined thresholds of discrete models. To investigate the performance of these models more systematically, we conducted simulations across a wide range of conditions, focusing on the comparative performance of the beta-density and step-function specifications and their robustness to misspecification of the selection function.

Simulation Methods

We conducted Monte Carlo simulation studies to assess the performance of the marginal beta-density selection model and to investigate whether the form of the assumed selection mechanism matters for estimation of average effect size. The simulations covered a wide range of conditions in which primary studies contributed multiple, statistically dependent effect size estimates. We compared estimates based on the beta density model to estimates generated from three existing methods: (1) a new version of the correlated hierarchical effects model with inverse sampling-covariance weights (CHE-ISCW), which accounts for dependency but not selective reporting (M. Chen & Pustejovsky, 2025); (2) the PET-PEESE method, which addresses selective reporting and has been adapted to handle dependent data structures (Stanley & Doucouliagos, 2014); and (3) a step-function selection model that accounts for both selective reporting and dependent effects (Pustejovsky, Citkowicz, et al., 2025). We evaluated the model estimates based on convergence rates, bias, accuracy, confidence interval coverage, and confidence interval width for estimating the average effect size of the unselected distribution. To ensure numerical stability in smaller samples, all selection models were estimated using the PML approach with the default penalties provided in the `metaselection` package (Pustejovsky, Joshi, et al., 2025). We evaluated bootstrap confidence interval performance under a narrower set of conditions to limit computational burden.

Simulations were conducted in R Version 4.4.0 (R Core Team, 2023) using the high-throughput computing cluster at the University of Wisconsin–Madison (Center for High Throughput Computing, 2006). The code relied on several R packages, including metafor (Viechtbauer, 2010), clubSandwich (Pustejovsky, 2024), simhelpers (Joshi & Pustejovsky, 2024), optimx (Nash & Varadhan, 2011), and tidyverse (Wickham et al., 2019).

Data generation

We generated simulated data using an approach similar to prior work (Pustejovsky, Citkowicz, et al., 2025), with the key difference that effect size estimates were selected for inclusion based on the beta-density selection model. For each simulated meta-analysis, we generated a pool of primary studies using a correlated-and-hierarchical effects model, with effect size estimates selected for inclusion according to probabilities defined by the beta-density selection model. Each study followed a two-group design, with sample sizes and numbers of effect sizes per study drawn from an empirical distribution based on the What

Works Clearinghouse database. We generated outcome correlations across studies by sampling from a beta distribution with mean ρ and standard deviation 0.05, then assumed a constant correlation between pairs of outcomes within a study.

Within each study, we simulated a study-level average effect size δ_j from a normal distribution with mean μ and variance τ_B^2, then generated individual effect size parameters from a normal distribution centered at δ_j with variance ω^2. Using these parameters, we drew multivariate normal outcomes for participants equally divided into treatment and control groups and computed standardized mean differences with Hedges's g small sample bias correction. One-sided p-values were computed for each effect size, and individual effect sizes were then selected for inclusion in the dataset, with selection probabilities determined by the beta-density selection model with parameters λ_1, λ_2. We repeated this process until the simulated meta-analytic dataset included at total of J studies with at least one observed result.

Estimation methods

We estimated the beta-density selection model using the proposed PML approach. This estimation utilized the package defaults, including a log-F penalty on the heterogeneity parameter and normal penalties on the selection parameters. We calculated cluster-robust standard errors using large-sample sandwich formulas. For a subset of simulation conditions, we also examined percentile confidence intervals based on the two-stage bootstrap. To maintain computational feasibility, we used $B = 399$ bootstrap replications of each estimator.

We compared the performance of the beta-density selection model to three other methods. First, we estimated a summary meta-analysis model using the CHE-ISCW approach (M. Chen & Pustejovsky, 2025), which accounts for effect size dependence but does not adjust for selective reporting. This method fits a CHE working model, but it allocates more weight to studies with smaller sampling variances by using generalized least squares with weighting matrices that are the inverse of the variance-covariance matrix of the sampling errors only. Following common practice in applied meta-analysis, we assumed a constant correlation of 0.80 within the working model, which aligns with the default correlation structure used in the `robumeta` package (Fisher, Tipton, & Zhang, 2024). This leads to a degree of misspecification when the average correlation used in the data-generating process differs from 0.80. We computed confidence intervals for the average effect size using cluster-robust variance estimation with the CR2 small-sample correction and Satterthwaite degrees of freedom.

Second, we estimated a variation of the PET/PEESE model originally proposed by Stanley and Doucouliagos (Stanley & Doucouliagos, 2014), adapted to handle dependent effect sizes. The PET model regresses effect size estimates on their standard errors, while the PEESE model uses sampling variances instead. Both models were estimated using the same working model assumptions as the CHE-ISCW model, including the assumed 0.80 correlation, with CR2 cluster-robust standard errors. Following established conventions (Stanley & Doucouliagos, 2014), we used the PET estimate if it was not statistically distinguishable from zero at an α-level of 0.10; otherwise, we used the PEESE estimate.

Third, we estimated the step-function selection model using the PML approach. As with the beta-density selection model, we used the default penalty structures for the step-function selection model as implemented in the `metaselection` package (Pustejovsky, Citkowicz, et al., 2025). We estimated two step-function selection models: (1) a one-step selection model with a single step at $\alpha_1 = 0.025$, and (2) a two-step selection model with steps at $\alpha_1 = 0.025$ and $\alpha_2 = 0.500$. Like the beta-density selection model, the one-step and two-step models are p-value selection models designed to address selective reporting, they are models for the marginal rather than joint distribution of estimates within studies, and they use cluster-robust variance estimation or bootstrapping to account for dependency. The main distinction between the selection models lies in the assumed functional form of the selection mechanism. Fitting the step-function models to data simulated under a beta-density

Table 2
Parameter values examined in the simulation study

Parameter	Full Simulation	Bootstrap Simulation
Overall average SMD (μ)	0.0, 0.2, 0.4, 0.8	0.0, 0.2, 0.4, 0.8
Between-study heterogeneity (τ)	0.05, 0.15, 0.30, 0.45	0.05, 0.30
Heterogeneity ratio (ω^2/τ^2)	0.0, 0.5	0.0, 0.5
Average correlation between outcomes (ρ)	0.40, 0.80	0.80
Probability of selection for non-affirmative effects (λ_1, λ_2)	(0.02, 0.90), (0.10, 0.90), (0.20, 0.90), (0.50, 0.90), (1.00, 1.00)	(0.20, 0.90), (0.50, 0.90), (1.00, 1.00)
Number of observed studies (J)	15, 30, 60, 90	15, 30, 60

selection process allows us to assess how robust these models are to misspecification of the selection function and whether the form of selection affects their performance.

Experimental design

We examined performance across a range of simulation conditions, summarized in Table 2. Manipulated parameters included overall average standardized mean difference (μ), between-study heterogeneity (τ_B), ratio of within- to between-study heterogeneity (ω^2/τ^2), average correlation between outcomes (ρ), probability of selection for non-affirmative results (λ_1, λ_2), and number of observed studies (J). The full simulation crossed all parameter values for a total of $4 \times 4 \times 2 \times 2 \times 5 \times 4 = 1,280$ conditions. For the more computationally intensive bootstrap simulations, we limited the design to a smaller subset of $4 \times 2 \times 2 \times 1 \times 3 \times 3 = 144$ conditions, focusing on smaller meta-analyses and reducing values for factors where results were stable (e.g., $\tau = 0.15$). For each condition, we generated 2,000 replications.

In the full simulation, we varied μ from 0.0 to 0.80, reflecting the range of effects observed in a large-scale review of education randomized controlled trials (Kraft, 2020), and τ from 0.05 (minimal heterogeneity) to 0.45 (substantial heterogeneity). Within-study heterogeneity was specified relative to between-study heterogeneity using a ratio of ω^2/τ^2

equal to 0 (no within-study heterogeneity) or 0.5.

To assess the impact of working model misspecification, we manipulated the average within-study correlation ρ across two levels: 0.80 (the default used in RVE software and correctly specified when $\rho = 0.80$) and 0.40 (inducing misspecification in the CHE-ISCW and PET-PEESE working models).

Selective reporting was modeled as the probability of selecting non-affirmative results based on the parameters of the beta-density. We considered five levels of selective reporting, including no selection ($\lambda_1 = 1.00, \lambda_2 = 1.00$), moderate selection ($\lambda_1 = 0.50, \lambda_2 = 0.90$), strong selection ($\lambda_1 = 0.20, \lambda_2 = 0.90$), very strong selection ($\lambda_1 = 0.10, \lambda_2 = 0.90$), and extreme selection ($\lambda_1 = 0.02, \lambda_2 = 0.90$).

We varied the number of observed studies (J) from 15 to 90, covering the typical size of meta-analyses in education and psychology (Tipton, Pustejovsky, & Ahmadi, 2019). We simulated primary study sample sizes based on the empirical distribution in the What Works Clearinghouse database. The sample sizes in the database ranged from 37 to 2,295 with a median of 211, and the number of effect sizes ranged from 1 to 48 with a median of 3.[1]

Performance criteria

We evaluated each method's performance in terms of convergence rates, bias, root mean-squared error (RMSE), 95% confidence interval coverage, and confidence interval width for the overall effect size μ. Bias reflects systematic deviation from the true parameter value, while RMSE captures both bias and sampling variability. For confidence intervals based on cluster-robust variance estimation, we defined coverage as the proportion of intervals that contained the true parameter value. For bootstrap intervals, we used $B = 399$ replicates per simulation due to computational limits—fewer than ideal for applied use. To estimate

[1] Prior work (Pustejovsky, Citkowicz, et al., 2025) included a condition in which the primary study sample sizes were divided by three to represent smaller studies, such as those in psychology lab settings. However, the simulation results for this condition were similar to those from the empirical distribution condition, so we have omitted it from the current simulations.

practical coverage rates, we followed an approach similar to Boos and Zhang (2000): we calculated coverage for smaller subsamples ($B = 49, 99, 199, 299$) randomly selected without replacement from the full set of $B = 399$ bootstraps, fit a regression of coverage on $1/B$, and used the intercept to extrapolate expected coverage for $B = 1999$. This extrapolation technique was previously evaluated and implemented (Pustejovsky, Citkowicz, et al., 2025, see Online Appendix C.3 for further details and demonstration of its performance).

Simulation Results

We organize our simulation results into two parts. First, we compare the performance of the beta-density selection model to the CHE-ISCW and PET/PEESE approaches. Specifically, we compare the bias and RMSE of the average effect size and the coverage rates of 95% confidence intervals. Second, we compare the performance of the beta-density selection model to the one-step and two-step selection models to assess the robustness of these models to misspecification of the selection function. Additional results concerning the bias and RMSE for the marginal variance of the effect size distribution for the selection models are provided in Online Appendix C and additional results on the performance of the selection parameters for the beta-density selection model are provided in Online Appendix D.

Sensitivity to Selective Reporting: Beta-Density vs. Standard Approaches

All three approaches—beta-density selection model, CHE-ISCW, and PET/PEESE—had perfect convergence rates, producing results for every replication across all $1,280$ conditions.

Bias

Figure 3 shows the bias for each method of estimating the average effect size (vertical axis) as a function of selective reporting strength (horizontal axis), average effect size (grid column), and between-study heterogeneity (τ, grid row). Each box plot summarizes variation in bias across the remaining simulation factors: the heterogeneity ratio, the correlation between effect size estimates, and the number of observed studies. Note that the vertical axis scale differs by grid row, reflecting how some methods' bias is sensitive to the

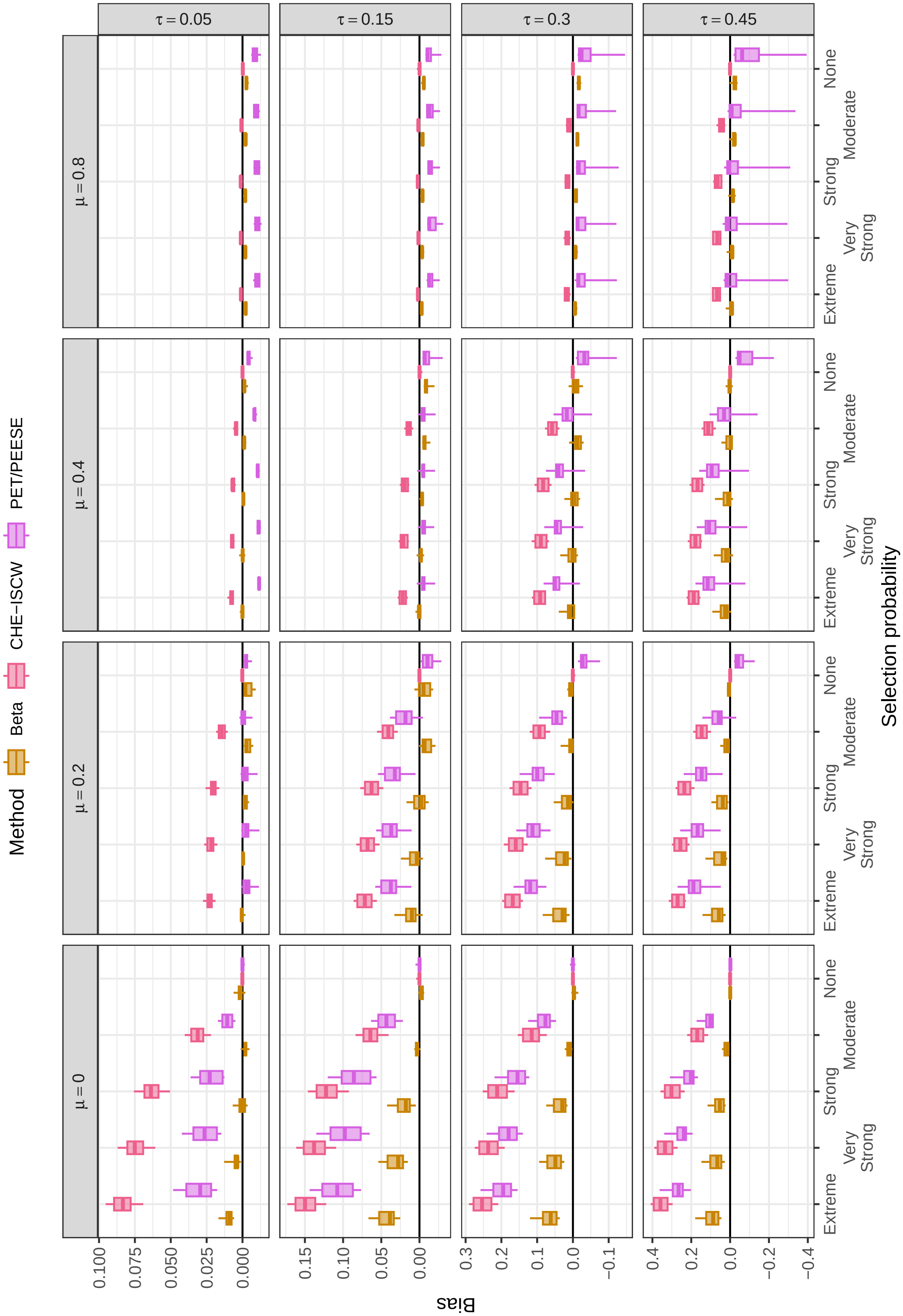

Figure 3

Bias of the average effect size by method, selection probability, average SMD, and between-study heterogeneity level of heterogeneity.

The beta-density selection model has negligible or small bias across all conditions, ranging from -0.04 to 0.18. Its bias is near or at zero for all conditions when the average effect size is non-zero or when selective reporting is moderate or absent. Bias is largest when average effect size is zero, selective reporting is very strong or extreme, and heterogeneity is moderate or large.

In contrast, bias for the comparison methods ranges from -0.01 to 0.41 for CHE-ISCW and -0.39 to 0.36 for PET/PEESE. Both methods are generally biased when selective reporting is present. For CHE-ISCW, which does not directly adjust for selective reporting, bias is closest to zero when average effect size is large ($\mu = 0.8$) and heterogeneity is low ($\tau \leq 0.15$). This result is not surprising because the large average effect size and low heterogeneity mean that the vast majority of simulated results are statistically significant, leaving little opportunity for selective reporting on the basis of statistical non-significance. For PET/PEESE, which uses a regression adjustment to account for possible selective reporting, bias is closest to zero when average effect size is moderate ($\mu = 0.4$) and heterogeneity is low ($\tau = 0.15$). For both comparison methods, bias grows stronger when selection is stronger, when average effect size is smaller, and when heterogeneity is larger. Bias is generally less pronounced for PET/PEESE than for CHE-ISCW.

Root Mean-Squared Error

RMSE captures both bias and variability, providing an overall measure of inaccuracy. Figure 4, constructed in the same format as Figure 3, shows the RMSE for each method of estimating the average effect size. Additional detail is provided in Figures B1 to B3 in Online Appendix B, which plot the ratio of RMSEs for each pair of methods to compare their relative accuracy.

Taken together, the figures indicate that no single method achieves the lowest RMSE

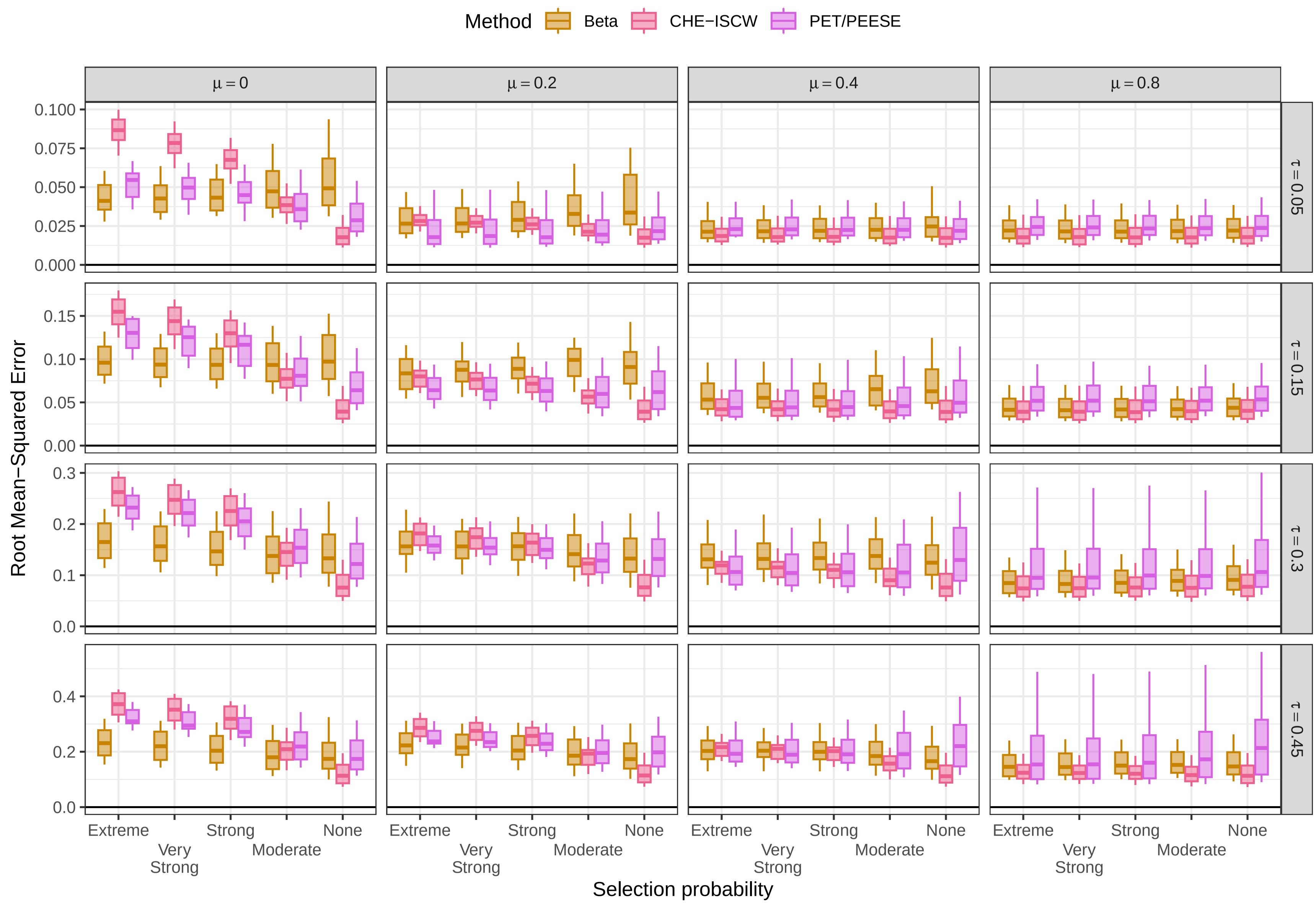

Figure 4

Root mean-squared error of the average effect size by method, selection probability, average SMD, and between-study heterogeneity across all conditions. Instead, each method reflects different bias–variance trade-offs. The beta-density selection model generally outperforms the others—achieving lower RMSE—when selective reporting is strong to extreme and $\mu = 0$ or when selective reporting is strong to extreme, $\mu = 0.2$, and heterogeneity is moderate to large ($\tau \geq 0.3$). CHE-ISCW consistently has the lowest RMSE when selective reporting is absent. PET/PEESE performs best when selective reporting is strong to extreme, $\mu = 0.2$, and heterogeneity is low ($\tau \leq 0.15$).

These bias–variance trade-offs stem from the fact that CHE-ISCW is more prone to bias when selection is present because it does not make an explicit adjustment for selective reporting. In contrast, the beta-density selection model tends to produce smaller biases under such conditions. However, when selective reporting is absent, CHE-ISCW is more precise than the methods that adjust for selective reporting (i.e., the beta-density selection model and PET/PEESE). Under those conditions, the added variability introduced by estimating a selection model or PET/PEESE adjustment outweighs the minimal reduction in bias they provide.

Confidence Interval Coverage

Figure 5 shows the coverage rates of 95% confidence intervals based on large-sample cluster-robust variance approximations for the three methods.[2] Across most conditions, coverage rates fall well below the nominal rate of 0.95 for all methods. Coverage rates range from 0.58 to 0.98 for the beta-density selection model, from 0 to 0.96 for CHE-ISCW, and from 0.02 to 0.96 for PET/PEESE. Coverage rates increase for all methods as the number of studies (J) increases, the average effect size (μ) increases, and heterogeneity (τ) decreases. When heterogeneity is small ($\tau \leq 0.15$) and the average effect size is moderate to large ($\mu \geq 0.4$), CHE-ISCW generally has the highest coverage rates and the beta-density selection

[2] Note that the vertical axis of Figure 5 is restricted to the range [0.5, 1.0], and coverage rates of the intervals based on CHE-ISCW and PET/PEESE are not depicted when they fall below 0.5. Figure B4 in Online Appendix B depicts the full range of coverage rates.

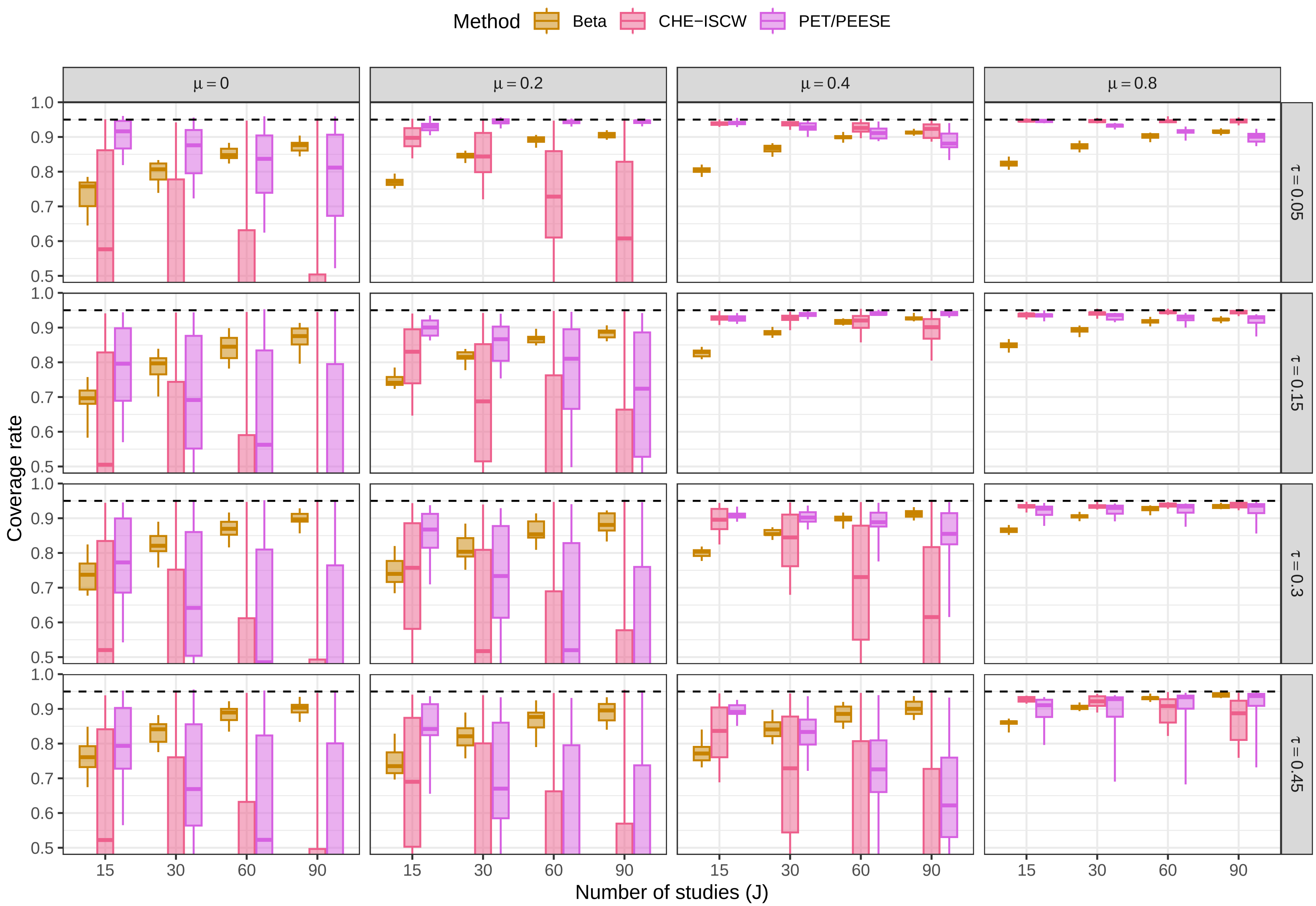

Figure 5

Coverage levels of confidence intervals for the average effect size based on cluster-robust variance approximations, by method, number of studies, average SMD, and between-study heterogeneity. Dashed lines correspond to the nominal confidence level of 0.95. Coverage rates of the CHE-ISCW and PET/PEESE intervals are not depicted when they fall below 0.5 model has the lowest coverage rates.

It is important to note that the confidence intervals produced by the comparison methods are often severely miscalibrated. When CHE-ISCW and PET/PEESE are biased due to selective reporting, their intervals tend to be centered away from the true parameter. As a result, as the number of studies increases, the standard errors of the average effect size estimate—and thus the interval widths—shrink, causing coverage rates to decline sharply, in some cases approaching zero. This occurs more frequently when heterogeneity is moderate to large ($\tau \geq 0.3$) and the average effect size is small ($\mu \leq 0.2$), particularly for CHE-ISCW.

Figure 6 depicts the coverage rates of 95% confidence intervals for the average effect size estimated using the beta-density selection model, comparing intervals based on the large-sample cluster-robust variance method and the two-stage percentile bootstrap method. Due to the computational demands of bootstrapping, we evaluated the bootstrap confidence intervals under a more limited range of data-generating conditions, including a maximum sample size of $J = 60$. Although neither method achieves exact nominal coverage across all conditions, the two-stage percentile bootstrap intervals consistently yield better coverage rates than those based on the large-sample cluster-robust variance method.

Robustness to Selection Function Misspecification: Beta-Density vs. Step Models

Across all 1,280 conditions, convergence rates were 100% for the beta-density selection model and over 99% for the one-step and two-step selection models.

Bias

Figure 7 displays the bias for the three selection models. Across most conditions, bias is consistently closer to zero when the average effect size is estimated using the beta-density selection model compared to the step-function selection models. Among the step-function selection models, the two-step model generally outperforms the one-step model, although both are prone to bias when the selection process is misspecified. The main exception occurs when average effect size is moderate to large ($\mu \geq 0.4$) and heterogeneity is low ($\tau \leq 0.15$),

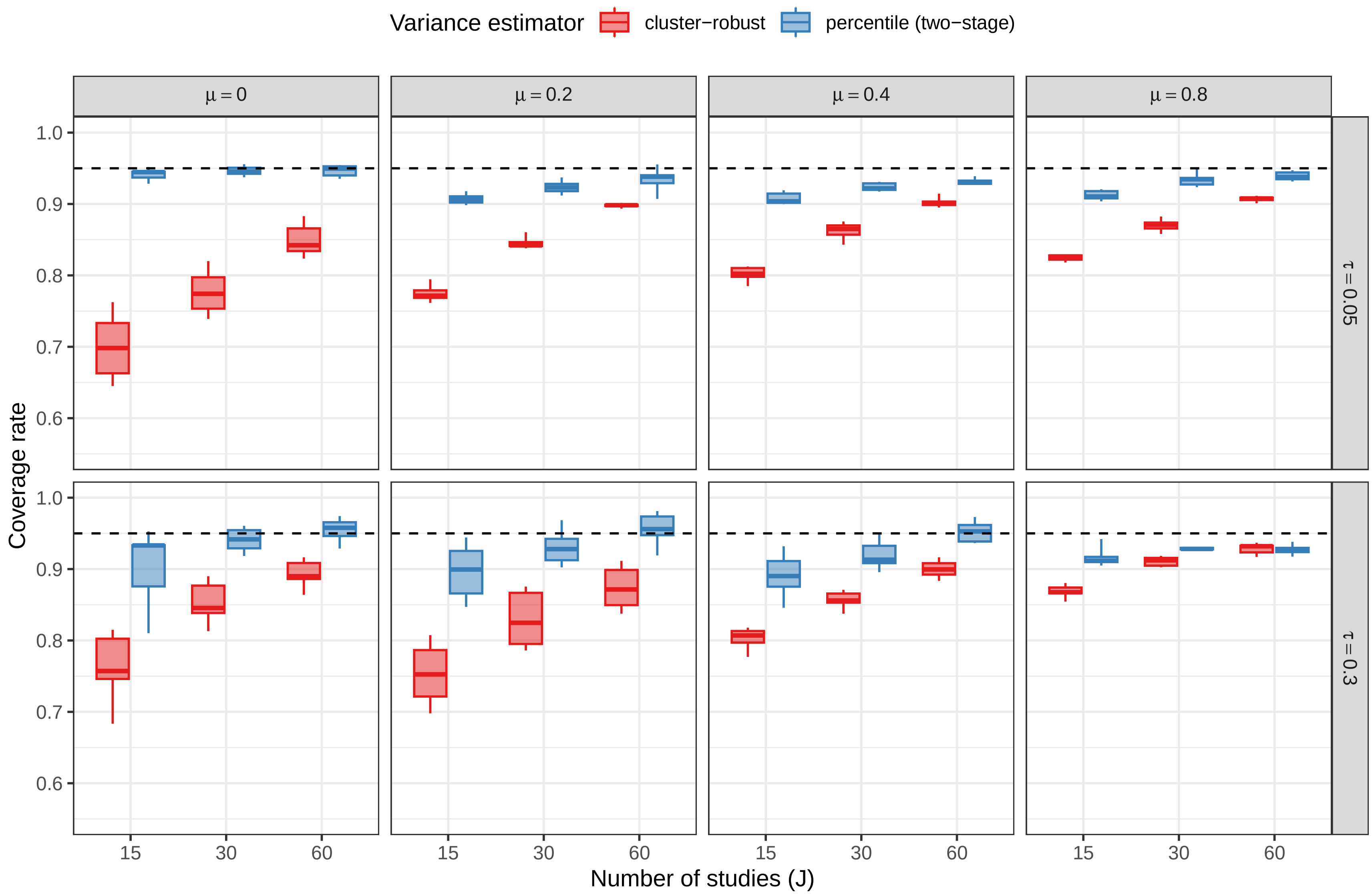

Figure 6

Coverage levels of confidence intervals for the average effect size estimated using the beta-density selection model, by method, number of studies, average SMD, and between-study heterogeneity. Dashed lines correspond to the nominal confidence level of 0.95.

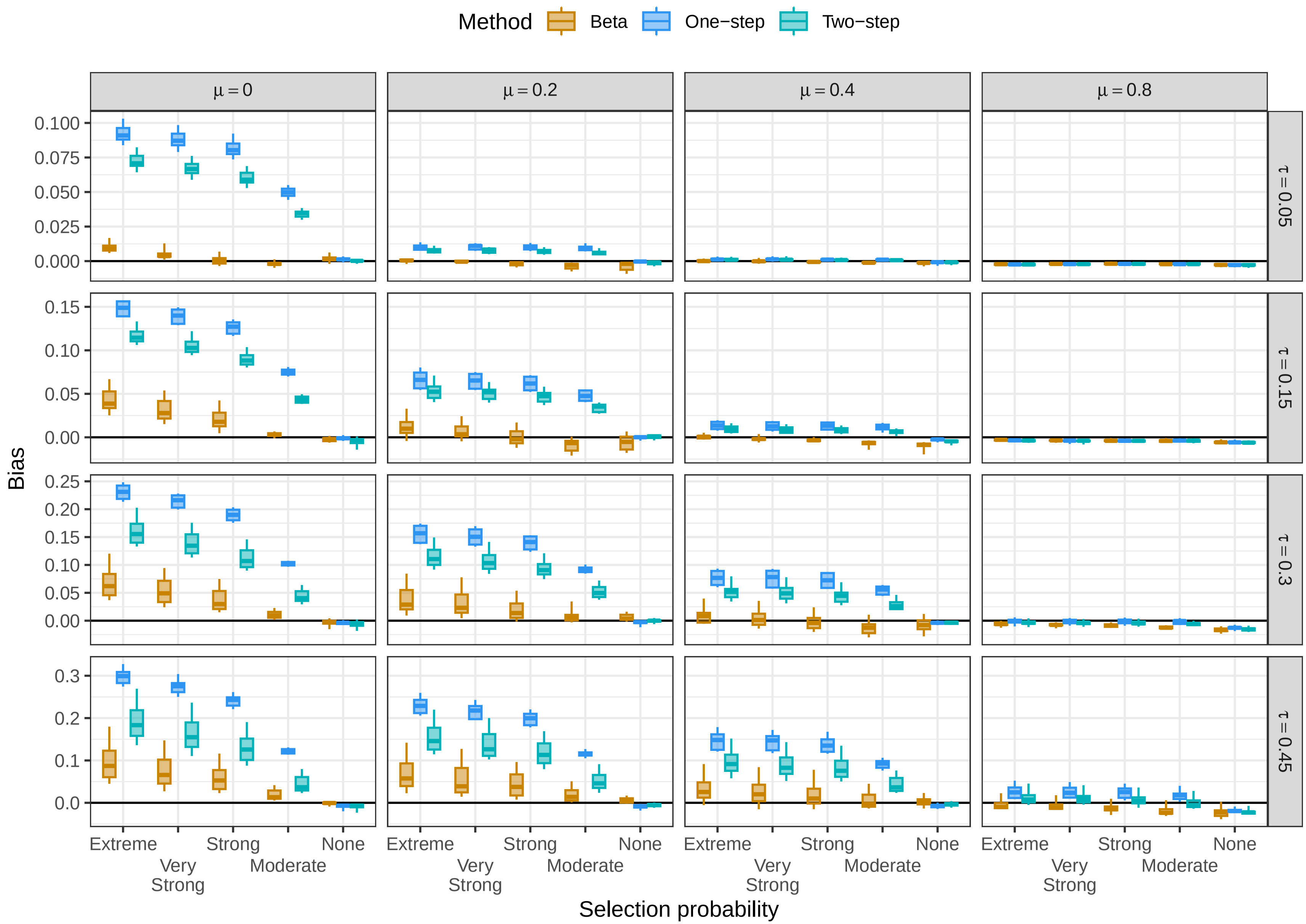

Figure 7

Bias of the average effect size by method, selection probability, average SMD, and between-study heterogeneity in which case all three models exhibit essentially zero bias. This pattern of biases is consistent with the fact that the data were generated under a beta-density selection process. The step-function selection models, particularly the one-step model, are not robust to misspecification of the selection mechanism under conditions where selective reporting produces meaningful bias.

Root Mean-Squared Error

Figure 8 presents the RMSE for the three selection models and highlights a trade-off between model complexity and precision. When $\mu = 0$ and selective reporting is strong to extreme, the results mirror the bias results pattern: the correctly specified beta-density selection model provides more accurate estimates than the step-function selection models, and the two-step model performs better than the one-step model. However, the relative performance shifts under other conditions. For instance, when $\mu = 0.4$, selective reporting is strong to extreme, and heterogeneity is moderate to large ($\tau \geq 0.30$), the two-step model yields estimates with lower RMSE than the beta-density model, even though the two-step model is misspecified. When average effect size is moderate or large ($\mu \geq 0.4$), selective reporting is absent to moderate, and heterogeneity is moderate to large ($\tau \geq 0.30$), the simpler, one-step model has a slight accuracy advantage over both alternatives, despite being misspecified. When $\mu = 0.2$, results are mixed. RMSE is similar across all three models when average effect size is moderate to large ($\mu \geq 0.4$) and heterogeneity is low ($\tau \leq 0.15$) because selective reporting produces little or no bias under these conditions.

Confidence Interval Coverage

Figure 9 shows the coverage rates of 95% confidence intervals based on two-stage bootstrap percentile intervals for the three models, with separate panels by level of average effect size (columns) and strength of selection (rows). Based on the results above, we focus on the coverage rates of the two-stage percentile bootstrap confidence intervals, as they consistently yielded coverage rates better than those from the large-sample cluster-robust variance method. Figure B5 in Online Appendix B depicts the coverage rates based on

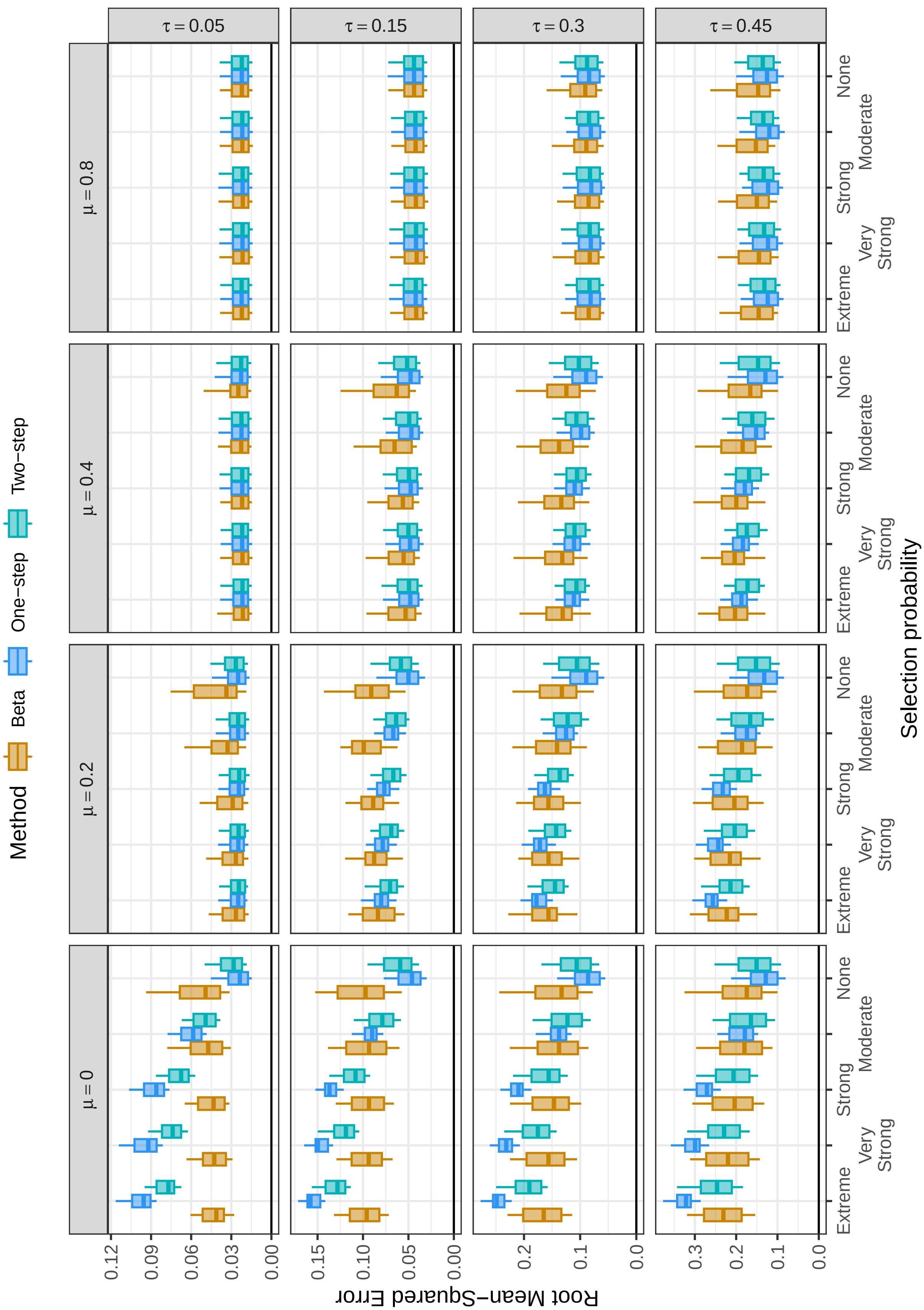

Figure 8

Root mean-squared error of the average effect size by method, selection probability, average SMD, and between-study heterogeneity

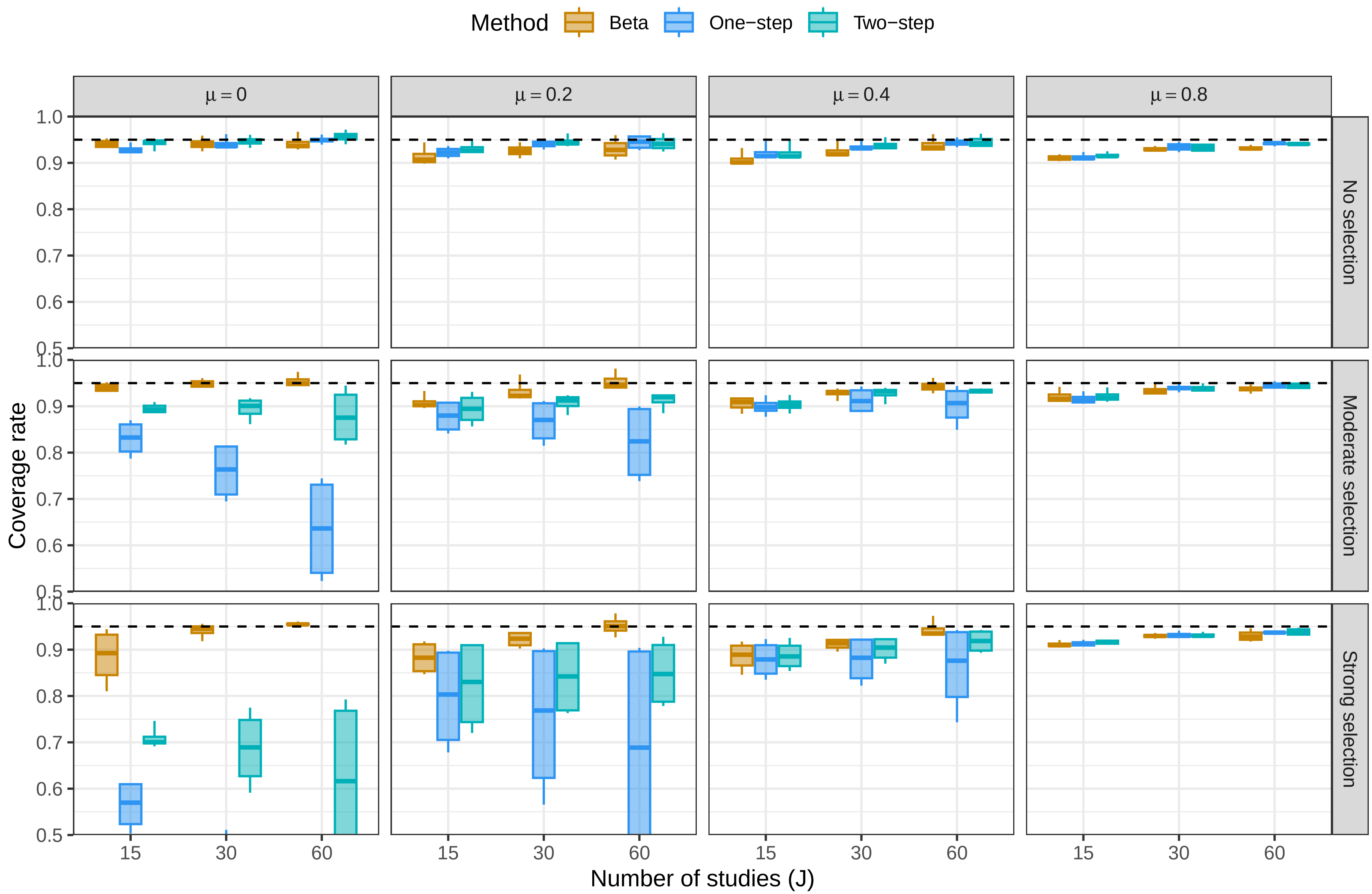

Figure 9

Coverage levels of confidence intervals for the average effect size based on two-stage bootstrap percentile intervals, by method, number of studies, average SMD, and strength of selective reporting. Dashed lines correspond to the nominal confidence level of 0.95.

cluster-robust variance approximations.

The beta-density selection model generally provides coverage levels that are closer to nominal than the two step-function models, with coverage levels improving towards the nominal level as the number of studies increases. The step-function selection models have lower coverage rates, particularly for stronger levels of selective reporting or when average effect size is smaller ($\mu \leq 0.2$). Of the two step-function selection models, coverage is generally better for the two-step model because of its greater flexibility to approximate the true selection process. However, the confidence intervals produced by both step-function selection models are sometimes very poorly calibrated and do not necessarily improve with larger sample sizes—especially under conditions of strong selection and small effect size where selective reporting leads to stronger bias.

Confidence Interval Width

Discussion

We have introduced and evaluated a PML approach for the beta-density selection model, providing a robust framework for adjusting for selective reporting in meta-analyses with dependent effect sizes. A persistent challenge in meta-analysis is the prevalence of dependent data structures——such as multiple outcomes or samples per study——which violate the independence assumptions of traditional selection models. To address this, we applied the marginal modeling strategy established in our previous work on the step-function selection model (Pustejovsky, Citkowicz, et al., 2025).

By modeling the *marginal* distribution of effect sizes and treating them as if they were independent observations for estimation purposes, we avoid the high-dimensional integration and complex joint-selection assumptions required by multivariate selection models. This “working independence” structure, paired with cluster-robust variance estimation or clustered bootstrapping, allows meta-analysts to account for the actual dependence in the data during the inference stage. Under this framework, the marginal model provides the point estimates, while RVE and bootstrapping provide valid inference by adjusting for the within-study correlations that the marginal model deliberately ignores.

While our earlier work demonstrated that this framework works well with step-function selection model, we recognized that this model imposes a somewhat rigid structure on the publication process. The step-function selection model assumes that selection is piecewise-constant, suggesting that the probability of selection only shifts at specific, discrete p-value thresholds. In reality, the relationship between statistical significance and reporting is likely more nuanced.

By incorporating the beta-density selection function (Citkowicz & Vevea, 2017) into our marginal modeling framework, we move toward a more adaptable representation of the selection process. The beta-density selection model offers a more flexible alternative for capturing diverse forms of selection that does not require meta-analysts to identify specific p-value thresholds a priori. Using only two selection parameters, this approach allows for a parsimonious representation of subtle, non-linear reporting patterns that a fixed-threshold model might overlook.

Our simulations indicate that, across all simulation conditions, the beta-density selection model demonstrated negligible to little bias and consistently outperformed CHE-ISCW and PET/PEESE in capturing the true average effect. However, results for the RMSE reveal a distinct bias–variance trade-off. While the beta-density model achieved the lowest RMSE under conditions of strong to extreme selection (specifically, when $\mu = 0$, or when $\mu = 0.2$ and heterogeneity was moderate to large), CHE-ISCW remained the most precise method—–yielding the lowest RMSE—–when selective reporting was absent. This occurs because the added variability introduced by estimating the beta-density selection model outweighs its minimal bias reduction in null-selection contexts. Regarding inference, while large-sample cluster-robust confidence intervals for all methods often fell below the nominal 95% rate—–particularly for CHE-ISCW and PET/PEESE as they centered on biased estimates—the beta-density model paired with the two-stage percentile bootstrap consistently provided the most reliable coverage.

The second part of our simulations focused on the robustness of selection models to functional misspecification, comparing the beta-density model to one-step and two-step selection models. Because the data were generated via a smooth beta-density process, the beta model naturally yielded bias closer to zero than the step-function alternatives, which proved less robust to the misspecification of a continuous selection mechanism. The step-function models—particularly the one-step variant—frequently produced biased estimates and miscalibrated, overly narrow confidence intervals. However, a notable bias–variance trade-off was observed: in certain conditions where the true average effect was moderate or large and selection was not extreme, the simpler one-step model occasionally yielded a lower RMSE than the beta-density model. This suggests that while the beta model is more flexible and recovers the functional form more accurately, the extreme parsimony of a misspecified step model can provide greater numerical stability in specific, low-information contexts.

Limitations and Future Directions

A primary limitation of the current study is the directionality of the misspecification tests. Because our simulation used the beta density as the "true" selection mechanism, the beta model held a natural specification advantage over the step-function selection models. A critical next step is to evaluate the "reverse" scenario: assessing the performance of the beta-density selection model when the true selection process is characterized by discrete, threshold-based jumps. Such an investigation would clarify whether the beta curve is flexible enough to approximate sharp discontinuities without introducing significant bias or artifactual patterns.

Furthermore, our focus on the marginal distribution of effect sizes assumes that selection operates primarily at the level of individual outcomes—what is often termed "selective outcome reporting." While our "working independence" approach using PML and RVE or bootstrapping handles the resulting data dependencies effectively, it does not explicitly model study-level selection (i.e., publication bias). In many contexts, an entire cluster of effects might be suppressed based on the significance of a single "headline" finding rather than the individual p-values of each outcome. Future work should explore the integration of study-level selection parameters into the marginal modeling framework to determine if this approach remains robust when selection occurs at multiple levels of the hierarchy.

Finally, a notable feature of selection models is their ability to incorporate predictors (moderators) to explain heterogeneity in the average effect. In the present study, we focused exclusively on the estimation of the marginal average effect size and did not include predictors in our simulation design. While the beta-density selection model is theoretically capable of adjusting for selective reporting while simultaneously estimating meta-regression coefficients, we did not test the model's performance—specifically its power to detect moderator effects or the accuracy of its standard errors—in the presence of covariates. Evaluating the model under meta-regression frameworks remains an important avenue for future research.

Conclusions

The beta-density selection model, estimated using a PML approach based on a composite marginal likelihood, offers a flexible and stable alternative for adjusting for selective reporting in meta-analyses with dependent data. By leveraging a marginal modeling strategy, it provides a bridge between the restrictive assumptions of the step-function selection model and the computational complexity of multivariate selection models. While our results highlight a bias–variance trade-off—where the beta model excels in high-selection contexts but may be less precise than simpler models when selection is absent—the beta-density selection model remained the most robust choice for maintaining

95% interval coverage when paired with the two-stage percentile bootstrap.

Given its performance in reducing bias—without requiring the a priori identification of thresholds—the beta-density selection model is a promising tool for sensitivity analysis. However, we must remain cautious; because the true selection mechanism in any given field is unknown and likely varies across disciplines, any choice of model involves inherent assumptions about the reporting process. The beta-density selection model does not claim to represent the "true" mechanism, but rather provides a more flexible functional form that can approximate a wide range of selection patterns. Pending further research into its performance under threshold-based selection, meta-regression contexts, and study-level publication bias, we recommend the beta-density selection model as a flexible option for meta-analysts. It is particularly well-suited for cases where researchers wish to move beyond a binary threshold approach and explore how sensitive their findings are to more nuanced, non-linear reporting patterns.

Author Contributions

MC: Conceptualization, Methodology, Investigation, Writing - original draft, Writing - review & editing, Project administration, Funding acquisition, Formal Analysis. **JEP:** Conceptualization, Methodology, Software, Validation, Investigation, Resources, Writing - original draft, Writing - review & editing, Formal Analysis.

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Data and Replication Materials

Code and data for replicating the empirical example and the Monte Carlo simulation study are available on the Open Science Framework at .

Conflict of Interest Statement

The authors declare no conflicts of interest.

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Appendix A

Technical details of cluster-robust variance estimation

This appendix provides further details about the penalized maximum likelihood (PML) estimation approach and cluster-robust variance estimation (CRVE) technique for the beta-density selection model. For estimation purposes, we parametrize the model using natural log transformations of the variance parameter and selection parameters, setting $\gamma = \log(\tau^2)$, $\zeta_1 = \log(\lambda_1)$, $\zeta_2 = \log(\lambda_2)$, and $\boldsymbol{\zeta} = (\zeta_1, \zeta_2)'$. We will denote the full parameter vector as $\boldsymbol{\theta} = (\boldsymbol{\beta}', \gamma, \boldsymbol{\zeta})'$. Let $w_{ij} = w(y_{ij}, \sigma_{ij}; \zeta_1, \zeta_2)$, $\mu_{ij} = \mathbf{x}_{ij}\boldsymbol{\beta}$, $\eta_{ij} = \exp(\gamma) + \sigma_{ij}^2$, and $A_{ij} = A(\mathbf{x}_{ij}, \sigma_{ij}; \boldsymbol{\beta}, \gamma, \boldsymbol{\zeta})$ as defined in Equation (5).

Equation (4) in the main text gives the marginal density of an observed effect size estimate under the beta-density model. Following from Equation (4), the log-likelihood contribution of effect i in study j is then

$$\begin{aligned} l_{ij}(\boldsymbol{\theta}) &= \log f(Y = y_{ij} | \sigma_{ij}, \mathbf{x}_{ij}) \\ &= \log w_{ij} - \log A_{ij} - \frac{1}{2}\frac{(y_{ij} - \mathbf{x}_{ij}\boldsymbol{\beta})^2}{\eta_{ij}} - \frac{1}{2}\log(\eta_{ij}) - \frac{1}{2}\log(2\pi), \end{aligned} \tag{A1}$$

The contribution of study j to the log-likelihood is then

$$l_j(\boldsymbol{\theta}) = \sum_{i=1}^{k_j} l_{ij}(\boldsymbol{\theta}) \tag{A2}$$

and the penalized marginal log-likelihood across all J studies is

$$l(\boldsymbol{\beta}, \gamma, \boldsymbol{\zeta}) = \sum_{j=1}^{J} l_j(\boldsymbol{\beta}, \gamma, \boldsymbol{\zeta}) + p_\beta(\boldsymbol{\beta}) + p_\gamma(\gamma) + p_{\boldsymbol{\zeta}}(\boldsymbol{\zeta}), \tag{A3}$$

where $p_\beta(\boldsymbol{\beta})$, $p_\gamma(\gamma)$, and $p_{\boldsymbol{\zeta}}(\boldsymbol{\zeta})$ are log-prior penalties for the meta-regression coefficients, log-variance, and log-selection parameters, respectively. The penalized marginal likelihood (PML) estimators, denoted as $\hat{\boldsymbol{\beta}}$, $\hat{\gamma}$, and $\hat{\boldsymbol{\zeta}}$, are obtained as the set of parameter values that maximize the penalized marginal log-likelihood for the observed data, as given in Equation (A3).

CRVE, or sandwich estimation, is a commonly used technique for quantifying the uncertainty with dependent observations where the dependence structure is not explicitly modeled. CRVE is defined in terms of the score contributions and Hessian matrix of the penalized log-likelihood. The score contribution of study j is the vector of derivatives of the penalized marginal log-likelihood contribution of study j:

$$\mathbf{S}_j\left(\boldsymbol{\theta}\right) = \frac{\partial l_j\left(\boldsymbol{\theta}\right)}{\partial \boldsymbol{\theta}}. \tag{A4}$$

The Hessian is the matrix of second derivatives of the penalized marginal log-likelihood, given by

$$\mathbf{H}(\boldsymbol{\theta}) = \sum_{j=1}^{J} \frac{\partial \mathbf{S}_j(\boldsymbol{\theta})}{\partial \theta'}. \tag{A5}$$

Let $\hat{\mathbf{S}}_j = \mathbf{S}_j(\hat{\boldsymbol{\beta}}, \hat{\gamma}, \hat{\boldsymbol{\zeta}})$ and $\hat{\mathbf{H}} = \mathbf{H}(\hat{\boldsymbol{\beta}}, \hat{\gamma}, \hat{\boldsymbol{\zeta}})$ denote the score contributions and Hessian matrix evaluated at the maximum of the penalized likelihood. A cluster-robust sandwich estimator of the sampling variance of the PML estimator is then

$$\mathbf{V}^{CR} = \hat{\mathbf{H}}^{-1}\left(\sum_{j=1}^{J} \hat{\mathbf{S}}_j \hat{\mathbf{S}}_j'\right)\hat{\mathbf{H}}^{-1}. \tag{A6}$$

We construct confidence intervals for model parameters using $\mathbf{V}^{CR}$ with Wald-type large sample approximations. For instance, the $(1-2\alpha)$-level large-sample confidence interval for a meta-regression parameter β_g is constructed as

$$\hat{\beta}_g \ \pm \ \Phi^{-1}(1-\alpha) \times \sqrt{V_{gg}^{CR}},$$

where V_{gg}^{CR} is the g^{th} diagonal entry of $\mathbf{V}^{CR}$.

It remains to give the exact form of the score contributions and Hessian matrix. To define these, observe that the beta-density selection function can be written compactly by defining

$$\tilde{p}_{ij} = \begin{cases} \alpha_1 & \text{if} \quad \Phi^{-1}(1-\alpha_1) \leq y_{ij}/\sigma_{ij} \\ \Phi\left(-y_{ij}/\sigma_{ij}\right) & \text{if} \quad \Phi^{-1}(1-\alpha_2) \leq y_{ij}/\sigma_{ij} < \Phi^{-1}(1-\alpha_1) \\ \alpha_2 & \text{if} \quad y_{ij}/\sigma_{ij} < \Phi^{-1}(1-\alpha_2), \end{cases} \tag{A7}$$

so that the selection weights of the truncated beta density model are

$$w_{ij} = w\left(y_{ij}, \sigma_{ij}; \zeta_1, \zeta_2\right) = \tilde{p}_{ij}^{\lambda_1 - 1}\left(1 - \tilde{p}_{ij}\right)^{\lambda_2 - 1}. \tag{A8}$$

Also let $c_{hij} = \eta_{ij}^{-1/2}\left(\sigma_{ij}\Phi^{-1}(1-\alpha_h) - \mu_{ij}\right)$ for $h = 1, 2$; $B_{0ij} = 1 - \Phi(c_{1ij})$; and $B_{2ij} = \Phi(c_{2ij})$.

The score and Hessian contributions of the truncated beta density model involve integrals that need to be calculated numerically. To ease notation for these integrals, we will denote the integral of a function $f(Y)$ with respect to the distribution of Y under the truncated beta density as

$$E_Y\left[f(Y)|\sigma_{ij}, \lambda_1, \lambda_2\right] = \int_{\sigma_{ij}\Phi^{-1}(1-\alpha_2)}^{\sigma_{ij}\Phi^{-1}(1-\alpha_1)} f(Y)\left[\Phi(-Y/\sigma_{ij})\right]^{\lambda_1}\left[\Phi(Y/\sigma_{ij})\right]^{\lambda_2} \frac{1}{\sqrt{\eta_{ij}}}\phi\left(\frac{Y - \mu_{ij}}{\sqrt{\eta_{ij}}}\right) dY. \tag{A9}$$

It follows that

$$A_{ij} = \alpha_1^{\lambda_1}(1-\alpha_1)^{\lambda_2} B_{0ij} + E_Y(1|\sigma_{ij}, \lambda_1, \lambda_2) + \alpha_2^{\lambda_1}(1-\alpha_2)^{\lambda_2} B_{2ij}. \tag{A10}$$

Score contributions

The components of the score contribution of study j involve derivatives of $l_j\left(\boldsymbol{\beta}, \gamma, \boldsymbol{\zeta}\right)$ with respect to all model parameters, as given by

$$\mathbf{S}_j\left(\boldsymbol{\beta}, \gamma, \boldsymbol{\zeta}\right) = \left(\mathbf{S}'_{\beta j}, S_{\gamma j}, \mathbf{S}'_{\zeta j}\right)'. \tag{A11}$$

The score contributions of study j have the general form

$$\mathbf{S}_{\beta j} = \sum_{i=1}^{k_j} \mathbf{x}'_{ij} \left(\frac{(y_{ij} - \mu_{ij})}{\eta_{ij}} - \frac{1}{A_{ij}} \frac{\partial A_{ij}}{\partial \mu_{ij}} \right) \tag{A12}$$

$$S_{\gamma j} = \sum_{i=1}^{k_j} \tau^2 \left(\frac{(y_{ij} - \mu_{ij})^2 - \eta_{ij}}{2\eta_{ij}^2} - \frac{1}{A_{ij}} \frac{\partial A_{ij}}{\partial \eta_{ij}} \right) \tag{A13}$$

$$S_{\zeta_h j} = \sum_{i=1}^{k_j} \lambda_h \left(\frac{1}{w_{ij}} \frac{\partial w_{ij}}{\partial \lambda_h} - \frac{1}{A_{ij}} \frac{\partial A_{ij}}{\partial \lambda_h} \right) \tag{A14}$$

with derivatives of w_{ij} given by

$$\frac{\partial w_{ij}}{\partial \lambda_1} = w_{ij} \log(\tilde{p}_{ij}) \tag{A15}$$

$$\frac{\partial w_{ij}}{\partial \lambda_2} = w_{ij} \log(1 - \tilde{p}_{ij}) \tag{A16}$$

and derivatives of A_{ij} given by

$$\frac{\partial A_{ij}}{\partial \mu_{ij}} = \frac{1}{\sqrt{\eta_{ij}}}\Bigg[\alpha_1^{\lambda_1}(1-\alpha_1)^{\lambda_2}\phi(c_{1ij}) \\ + E_Y\left(\frac{Y-\mu_{ij}}{\sqrt{\eta_{ij}}}\,\middle|\,\sigma_{ij},\lambda_1,\lambda_2\right) - \alpha_2^{\lambda_1}(1-\alpha_2)^{\lambda_2}\phi(c_{2ij})\Bigg] \tag{A17}$$

$$\frac{\partial A_{ij}}{\partial \eta_{ij}} = \frac{1}{2\eta_{ij}}\Bigg[\alpha_1^{\lambda_1}(1-\alpha_1)^{\lambda_2}c_{1ij}\phi(c_{1ij}) \\ + E_Y\left(\frac{(Y-\mu_{ij})^2}{\eta_{ij}} - 1\,\middle|\,\sigma_{ij},\lambda_1,\lambda_2\right) - \alpha_2^{\lambda_1}(1-\alpha_2)^{\lambda_2}c_{2ij}\phi(c_{2ij})\Bigg] \tag{A18}$$

$$\frac{\partial A_{ij}}{\partial \lambda_1} = \log(\alpha_1)\alpha_1^{\lambda_1}(1-\alpha_1)^{\lambda_2}B_{0ij} \\ + E_Y\left[\log\Phi(-Y/\sigma_{ij}) \mid \sigma_{ij},\lambda_1,\lambda_2\right] + \log(\alpha_2)\alpha_2^{\lambda_1}(1-\alpha_2)^{\lambda_2}B_{2ij} \tag{A19}$$

$$\frac{\partial A_{ij}}{\partial \lambda_2} = \log(1-\alpha_1)\alpha_1^{\lambda_1}(1-\alpha_1)^{\lambda_2}B_{0ij} \\ + E_Y\left[\log\Phi(Y/\sigma_{ij}) \mid \sigma_{ij},\lambda_1,\lambda_2\right] + \log(1-\alpha_2)\alpha_2^{\lambda_1}(1-\alpha_2)^{\lambda_2}B_{2ij}. \tag{A20}$$

Hessian

The sub-components of the Hessian matrix have the following forms:

$$\mathbf{H}^{\beta\beta'} = \sum_{j=1}^{J}\sum_{i=1}^{k_j} \mathbf{x}'_{ij}\mathbf{x}_{ij} \left(\frac{1}{A_{ij}^2} \left(\frac{\partial A_{ij}}{\partial \mu_{ij}} \right)^2 - \frac{1}{A_{ij}} \frac{\partial^2 A_{ij}}{\partial \mu_{ij}^2} - \frac{1}{\eta_{ij}} \right) \tag{A21}$$

$$\mathbf{H}^{\beta\gamma} = \sum_{j=1}^{J}\sum_{i=1}^{k_j} \mathbf{x}'_{ij}\tau^2 \left(\frac{1}{A_{ij}^2} \frac{\partial A_{ij}}{\partial \mu_{ij}} \frac{\partial A_{ij}}{\partial \eta_{ij}} - \frac{1}{A_{ij}} \frac{\partial^2 A_{ij}}{\partial \mu_{ij}\partial \eta_{ij}} - \frac{y_{ij} - \mu_{ij}}{\eta_{ij}^2} \right) \tag{A22}$$

$$\mathbf{H}^{\beta\zeta_h} = \sum_{j=1}^{J}\sum_{i=1}^{k_j} \mathbf{x}'_{ij}\lambda_h \left(\frac{1}{A_{ij}^2} \frac{\partial A_{ij}}{\partial \mu_{ij}} \frac{\partial A_{ij}}{\partial \lambda_h} - \frac{1}{A_{ij}} \frac{\partial^2 A_{ij}}{\partial \mu_{ij}\partial \lambda_h} \right) \tag{A23}$$

$$\begin{aligned}\mathbf{H}^{\gamma\gamma} = \sum_{j=1}^{J}\sum_{i=1}^{k_j} \Bigg[&\tau^2 \left(\frac{(y_{ij} - \mu_{ij})^2 - \eta_{ij}}{2\eta_{ij}^2} - \frac{1}{A_{ij}} \frac{\partial A_{ij}}{\partial \eta_{ij}} \right) \\ &+ \tau^4 \left(\frac{1}{A_{ij}^2} \left(\frac{\partial A_{ij}}{\partial \eta_{ij}} \right)^2 - \frac{1}{A_{ij}} \frac{\partial^2 A_{ij}}{\partial \eta_{ij}^2} - \frac{2(y_{ij} - \mu_{ij})^2 - \eta_{ij}}{2\eta_{ij}^3} \right) \Bigg]\end{aligned} \tag{A24}$$

$$\mathbf{H}^{\gamma\zeta_h} = \sum_{j=1}^{J}\sum_{i=1}^{k_j} \tau^2\lambda_h \left(\frac{1}{A_{ij}^2} \frac{\partial A_{ij}}{\partial \eta_{ij}} \frac{\partial A_{ij}}{\partial \lambda_h} - \frac{1}{A_{ij}} \frac{\partial^2 A_{ij}}{\partial \eta_{ij}\partial \lambda_h} \right) \tag{A25}$$

$$\begin{aligned}\mathbf{H}^{\zeta_f\zeta_h} = \sum_{j=1}^{J}\sum_{i=1}^{k_j} \lambda_f\lambda_h &\left(\frac{1}{w_{ij}} \frac{\partial^2 w_{ij}}{\partial \lambda_f \partial \lambda_h} - \frac{1}{w_{ij}^2} \frac{\partial w_{ij}}{\partial \lambda_f} \frac{\partial w_{ij}}{\partial \lambda_h} + \frac{1}{A_{ij}^2} \frac{\partial A_{ij}}{\partial \lambda_f} \frac{\partial A_{ij}}{\partial \lambda_h} - \frac{1}{A_{ij}} \frac{\partial^2 A_{ij}}{\partial \lambda_f \partial \lambda_h} \right) \\ &+ I(f = h)\lambda_f \left(\frac{1}{w_{ij}} \frac{\partial w_{ij}}{\partial \lambda_f} - \frac{1}{A_{ij}} \frac{\partial A_{ij}}{\partial \lambda_f} \right),\end{aligned} \tag{A26}$$

The second derivatives of the weight function with respect to λ_1 and λ_2 are

$$\frac{\partial^2 w_{ij}}{\partial \lambda_1^2} = w_{ij} \left[\log(\tilde{p}_{ij}) \right]^2 \tag{A27}$$

$$\frac{\partial^2 w_{ij}}{\partial \lambda_2^2} = w_{ij} \left[\log(1 - \tilde{p}_{ij}) \right]^2 \tag{A28}$$

$$\frac{\partial^2 w_{ij}}{\partial \lambda_1 \partial \lambda_2} = w_{ij} \log(\tilde{p}_{ij}) \log(1 - \tilde{p}_{ij}). \tag{A29}$$

The second derivatives of A_{ij} are

$$\begin{aligned}\frac{\partial^2 A_{ij}}{\partial \mu_{ij}^2} = \frac{1}{\eta_{ij}} \Bigg[&\alpha_1^{\lambda_1}(1 - \alpha_1)^{\lambda_2} c_{1ij}\phi(c_{1ij}) \\ &+ E_Y \left(\frac{(Y - \mu_{ij})^2}{\eta_{ij}} - 1 \,\middle|\, \sigma_{ij}, \lambda_1, \lambda_2 \right) - \alpha_2^{\lambda_1}(1 - \alpha_2)^{\lambda_2} c_{2ij}\phi(c_{2ij}) \Bigg]\end{aligned} \tag{A30}$$

$$\frac{\partial^2 A_{ij}}{\partial \mu_{ij} \partial \eta_{ij}} = \frac{1}{2\eta_{ij}^{3/2}} \Bigg[\alpha_1^{\lambda_1}(1-\alpha_1)^{\lambda_2}(c_{1ij}^2 - 1)\phi(c_{1ij}) + E_Y\left(\frac{(Y-\mu_{ij})[(Y-\mu_{ij})^2 - 3\eta_{ij}]}{\eta_{ij}^{3/2}} \,\middle|\, \sigma_{ij}, \lambda_1, \lambda_2 \right) - \alpha_2^{\lambda_1}(1-\alpha_2)^{\lambda_2}(c_{2ij}^2 - 1)\phi(c_{2ij}) \Bigg] \tag{A31}$$

$$\frac{\partial^2 A_{ij}}{\partial \mu_{ij} \partial \lambda_1} = \frac{1}{\sqrt{\eta_{ij}}} \Bigg[\log(\alpha_1)\alpha_1^{\lambda_1}(1-\alpha_1)^{\lambda_2}\phi(c_{1ij}) + E_Y\left(\frac{Y-\mu_{ij}}{\sqrt{\eta_{ij}}} \times \log\Phi(-Y/\sigma_{ij}) \,\middle|\, \sigma_{ij}, \lambda_1, \lambda_2 \right) - \log(\alpha_2)\alpha_2^{\lambda_1}(1-\alpha_2)^{\lambda_2}\phi(c_{2ij}) \Bigg] \tag{A32}$$

$$\frac{\partial^2 A_{ij}}{\partial \mu_{ij} \partial \lambda_2} = \frac{1}{\sqrt{\eta_{ij}}} \Bigg[\log(1-\alpha_1)\alpha_1^{\lambda_1}(1-\alpha_1)^{\lambda_2}\phi(c_{1ij}) + E_Y\left(\frac{Y-\mu_{ij}}{\sqrt{\eta_{ij}}} \times \log\Phi(Y/\sigma_{ij}) \,\middle|\, \sigma_{ij}, \lambda_1, \lambda_2 \right) - \log(1-\alpha_2)\alpha_2^{\lambda_1}(1-\alpha_2)^{\lambda_2}\phi(c_{2ij}) \Bigg] \tag{A33}$$

$$\frac{\partial^2 A_{ij}}{\partial \eta_{ij}^2} = \frac{1}{4\eta_{ij}^2} \Bigg[\alpha_1^{\lambda_1}(1-\alpha_1)^{\lambda_2}(c_{1ij}^3 - 3c_{1ij})\phi(c_{1ij}) + E_Y\left(\frac{(Y-\mu_{ij})^4}{\eta_{ij}^2} - 6\frac{(Y-\mu_{ij})^2}{\eta_{ij}} + 3 \,\middle|\, \sigma_{ij}, \lambda_1, \lambda_2 \right) - \alpha_2^{\lambda_1}(1-\alpha_2)^{\lambda_2}(c_{2ij}^3 - 3c_{2ij})\phi(c_{2ij}) \Bigg] \tag{A34}$$

$$\begin{aligned}\frac{\partial^2 A_{ij}}{\partial\eta_{ij}\partial\lambda_1} = \frac{1}{2\eta_{ij}}\Bigg[&\log(\alpha_1)\alpha_1^{\lambda_1}(1-\alpha_1)^{\lambda_2}c_{1ij}\phi(c_{1ij})\\ &+E_Y\left[\log\Phi(-Y/\sigma_{ij})\times\left(\frac{(Y-\mu_{ij})^2}{\eta_{ij}}-1\right)\,\Big|\,\sigma_{ij},\lambda_1,\lambda_2\right]\\ &-\log(\alpha_2)\alpha_2^{\lambda_1}(1-\alpha_2)^{\lambda_2}c_{2ij}\phi(c_{2ij})\Bigg]\end{aligned} \tag{A35}$$

$$\begin{aligned}\frac{\partial^2 A_{ij}}{\partial\eta_{ij}\partial\lambda_2} = \frac{1}{2\eta_{ij}}\Bigg[&\log(1-\alpha_1)\alpha_1^{\lambda_1}(1-\alpha_1)^{\lambda_2}c_{1ij}\phi(c_{1ij})\\ &+E_Y\left[\log\Phi(Y/\sigma_{ij})\times\left(\frac{(Y-\mu_{ij})^2}{\eta_{ij}}-1\right)\,\Big|\,\sigma_{ij},\lambda_1,\lambda_2\right]\\ &-\log(1-\alpha_2)\alpha_2^{\lambda_1}(1-\alpha_2)^{\lambda_2}c_{2ij}\phi(c_{2ij})\Bigg]\end{aligned} \tag{A36}$$

$$\begin{aligned}\frac{\partial^2 A_{ij}}{\partial\lambda_1^2} &= [\log(\alpha_1)]^2\,\alpha_1^{\lambda_1}(1-\alpha_1)^{\lambda_2}B_{0ij}\\ &\quad+E_Y\left[(\log\Phi(-Y/\sigma_{ij}))^2\,\Big|\,\sigma_{ij},\lambda_1,\lambda_2\right]\\ &\quad\quad+(\log(\alpha_2))^2\,\alpha_2^{\lambda_1}(1-\alpha_2)^{\lambda_2}B_{2ij}\end{aligned} \tag{A37}$$

$$\begin{aligned}\frac{\partial^2 A_{ij}}{\partial\lambda_2^2} &= [\log(1-\alpha_1)]^2\,\alpha_1^{\lambda_1}(1-\alpha_1)^{\lambda_2}B_{0ij}\\ &\quad+E_Y\left[(\log\Phi(Y/\sigma_{ij}))^2\,\Big|\,\sigma_{ij},\lambda_1,\lambda_2\right]\\ &\quad\quad+(\log(1-\alpha_2))^2\,\alpha_2^{\lambda_1}(1-\alpha_2)^{\lambda_2}B_{2ij}\end{aligned} \tag{A38}$$

$$\begin{aligned}\frac{\partial^2 A_{ij}}{\partial\lambda_1\partial\lambda_2} &= \log(\alpha_1)\log(1-\alpha_1)\alpha_1^{\lambda_1}(1-\alpha_1)^{\lambda_2}B_{0ij}\\ &\quad+E_Y\left[\log\Phi(-Y/\sigma_{ij})\log\Phi(Y/\sigma_{ij})\,|\,\sigma_{ij},\lambda_1,\lambda_2\right]\\ &\quad\quad+\log(\alpha_2)\log(1-\alpha_2)\alpha_2^{\lambda_1}(1-\alpha_2)^{\lambda_2}B_{2ij}\end{aligned} \tag{A39}$$

Appendix B

Additional simulation results for methods of estimating average effect size (μ)

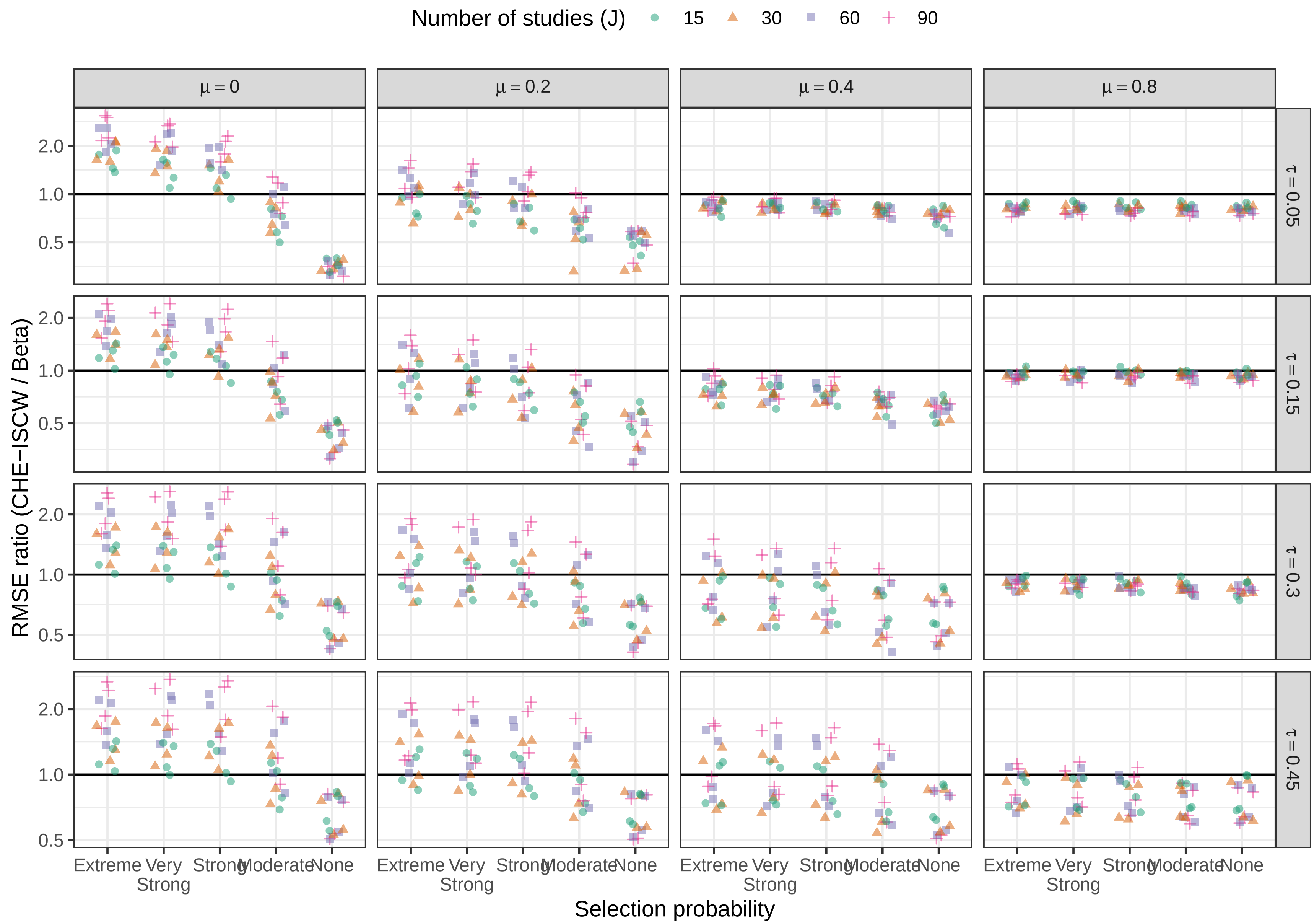

Figure B1

Ratio of root mean-squared error of the average effect for CHE-ISCW to root mean-squared error of beta-density by selection probability, number of studies, average SMD, and between-study heterogeneity

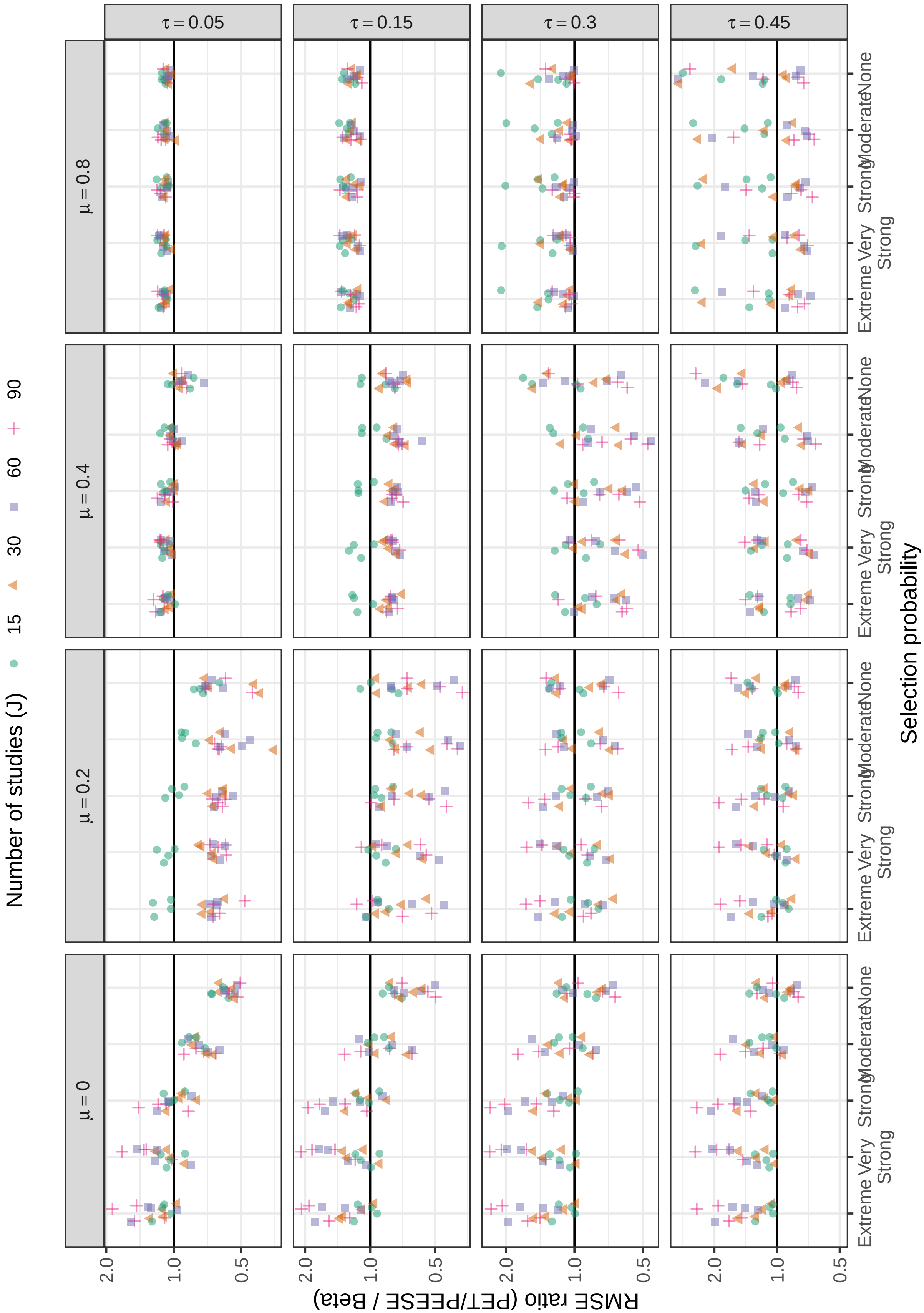

Figure B2

Ratio of root mean-squared error of the average effect for PET/PEESE to root mean-squared error of beta-density by selection probability, number of studies, average SMD, and between-study heterogeneity

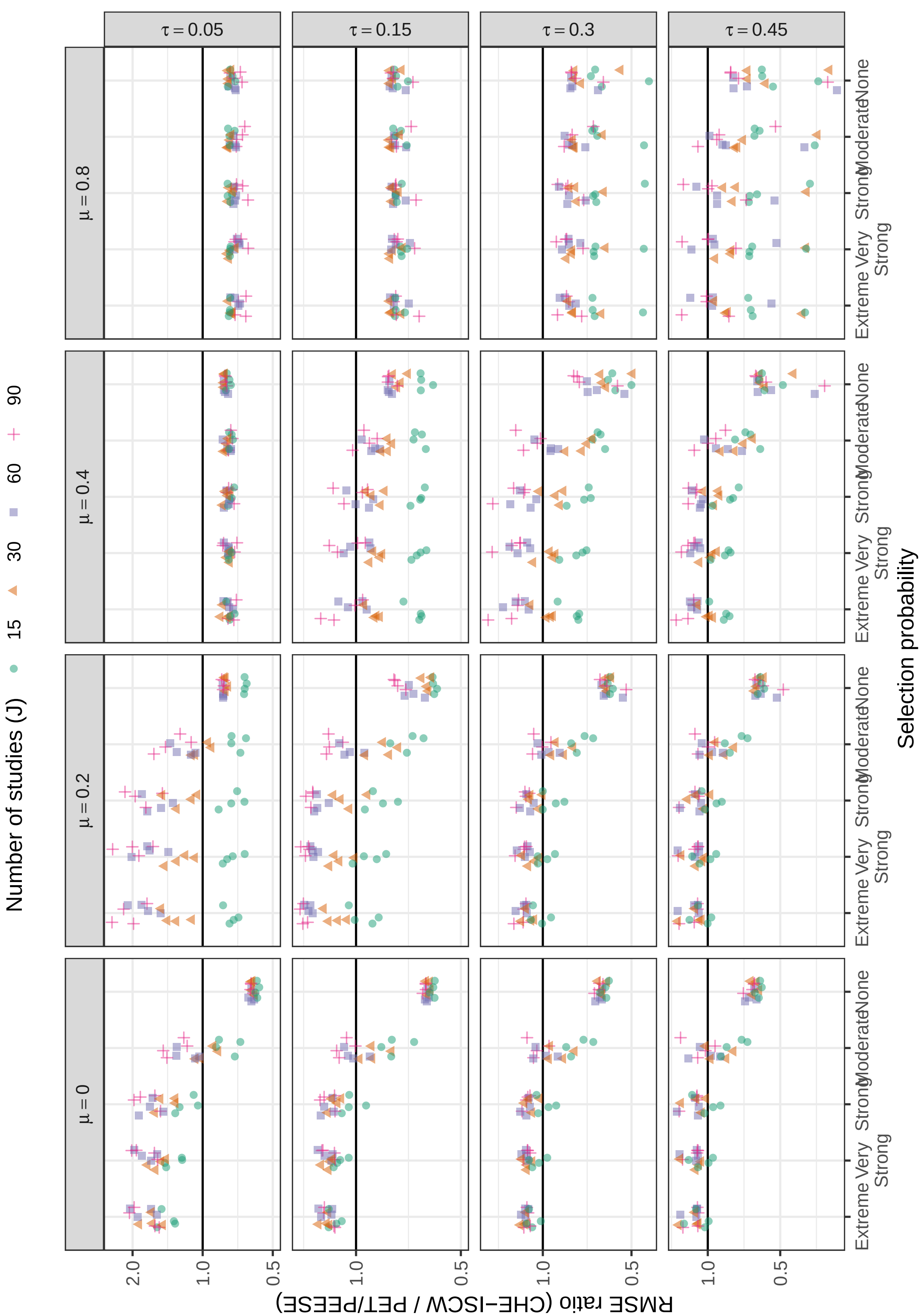

Figure B3

Ratio of root mean-squared error of the average effect for CHE-ISCW to root mean-squared error of PET/PEESE by selection probability, number of studies, average SMD, and between-study heterogeneity

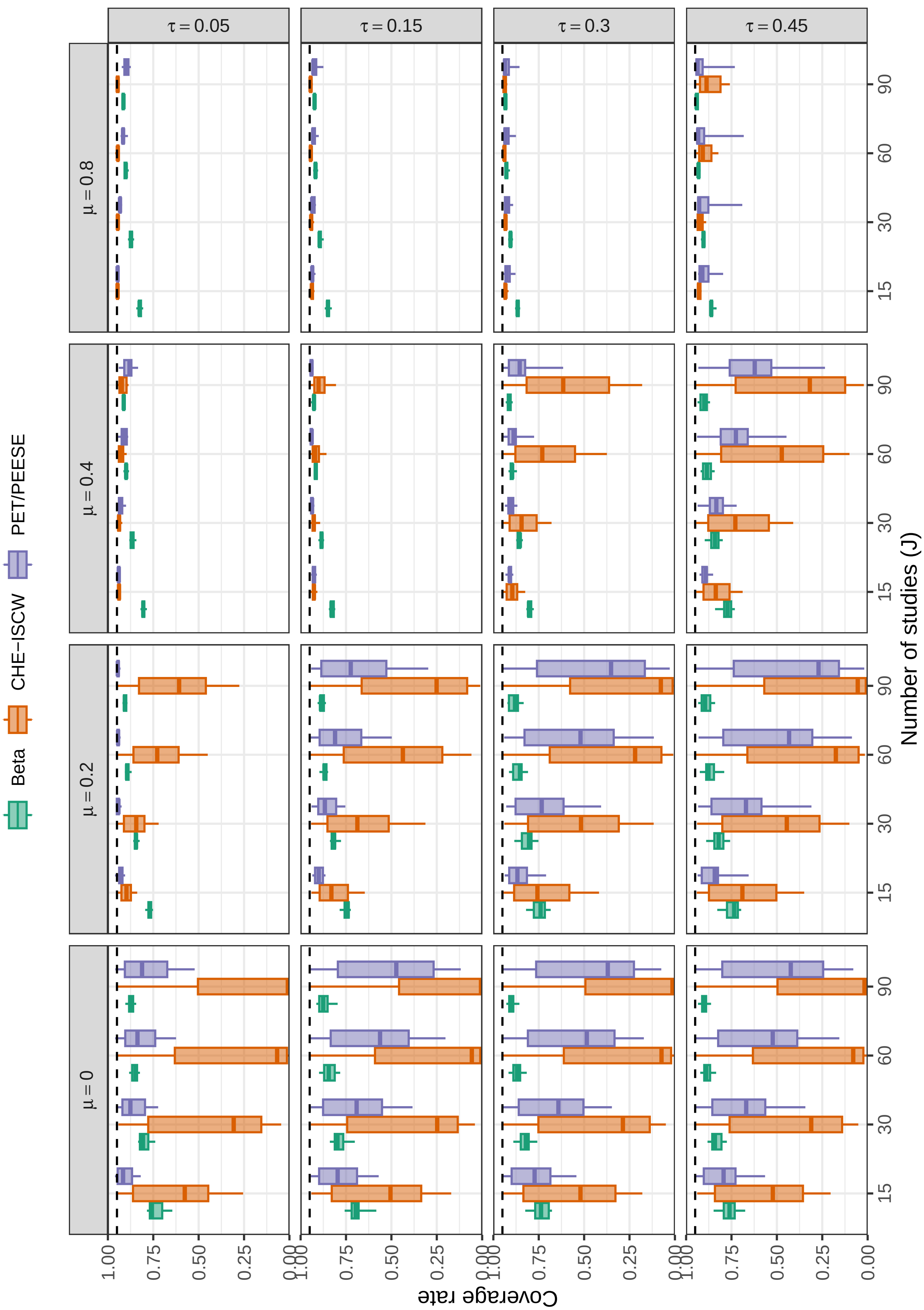

Figure B4

Coverage levels of confidence intervals for the average effect size based on cluster-robust variance approximations, by method, number of studies, average SMD, and between-study heterogeneity. Dashed lines correspond to the nominal confidence level of 0.95.

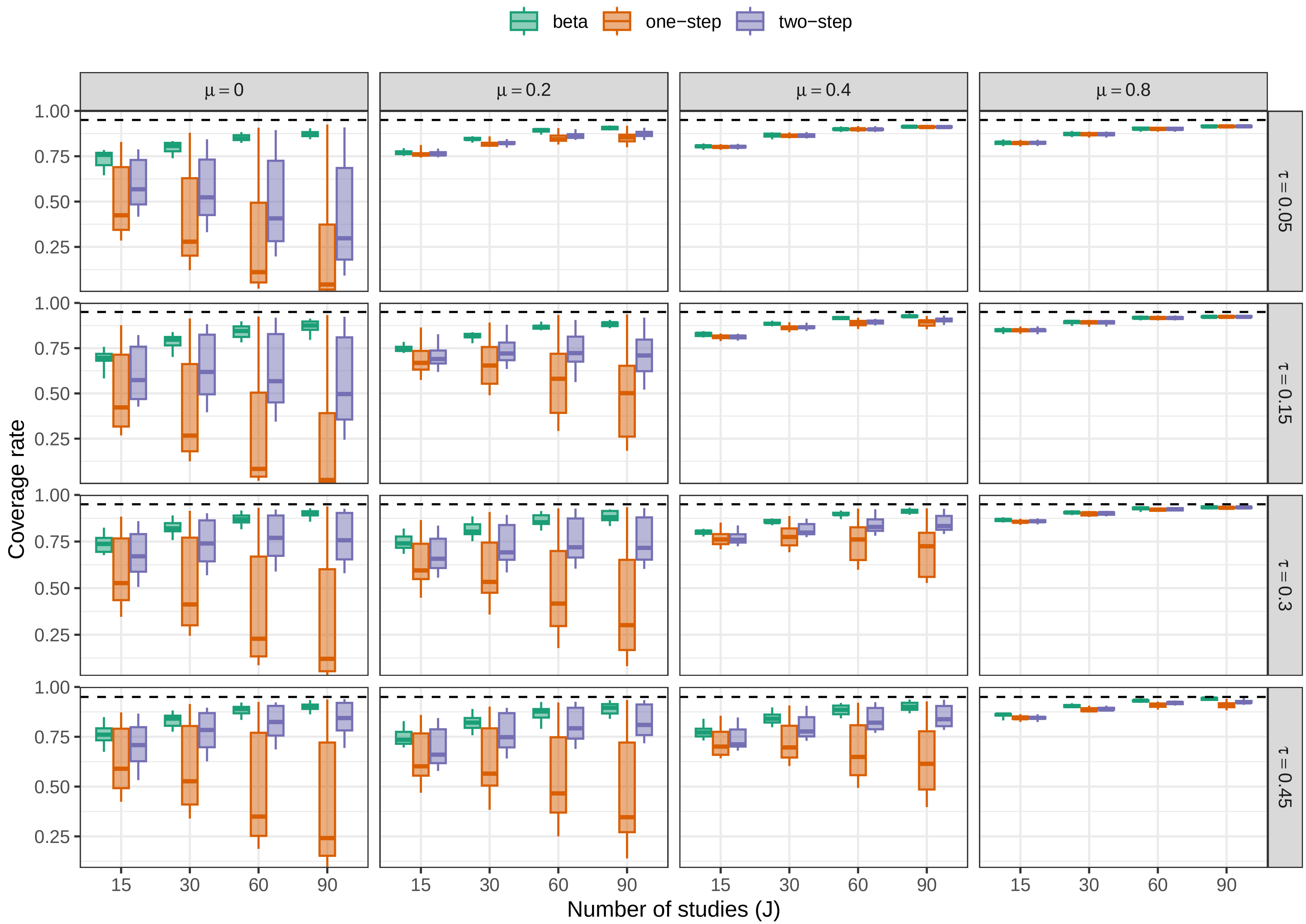

Figure B5

Coverage levels of confidence intervals for the average effect size based on cluster-robust variance approximations, by method, number of studies, average SMD, and between-study heterogeneity. Dashed lines correspond to the nominal confidence level of 0.95.

Appendix C

Additional simulation results for methods of estimating heterogeneity

We briefly examine the estimation of the marginal heterogeneity of the effect size distribution. Figure C1 shows the relative bias of marginal variance $\tau_B^2 + \omega^2$ for the three selection models: beta-density, one-step, and two-step. In most conditions, all three models are biased in the negative direction, with relative bias consistently below zero when between-study heterogeneity is $\tau_B \geq 0.15$. The beta-density selection model generally has relative bias closest to zero, particularly when selective reporting is present (moderate to extreme). Among the step-function selection models, the two-step model generally outperforms the one-step model. Except the one-step model generally outperforms both the beta-density and two-step models when average effect size is small ($\mu < 0.2$) and selective reporting is not present, though the difference is minimal. Relative bias is very similar across the three models when average effect size is large ($\mu = 0.8$) and between-study heterogeneity is small ($\tau_B \leq 0.15$).

Figure C2 depicts the relative RMSE of the marginal heterogeneity for the three selection models, with Figures C3 to C5 plotting the ratio of RMSEs for each pair of models to compare their relative accuracy. The beta-density selection model is the most accurate (i.e., has the lowest RMSE) when selective reporting is present, average effect size is small ($\mu < 0.2$), and between-study heterogeneity is moderate to large ($\tau_B \geq 0.30$). A bias-variance trade-off takes places when $\mu = 0.4$, with the one-step model outperforming the others when bias is not present and the two-step model outperforming the others when bias is present. The one-step model also outperforms the others when average effect size is large ($\mu = 0.8$) and between-study heterogeneity is moderate to large ($\tau_B \geq 0.30$). RMSE is very similar across the three models when average effect size is large ($\mu = 0.8$) and between-study heterogeneity is small ($\tau_B \leq 0.15$).

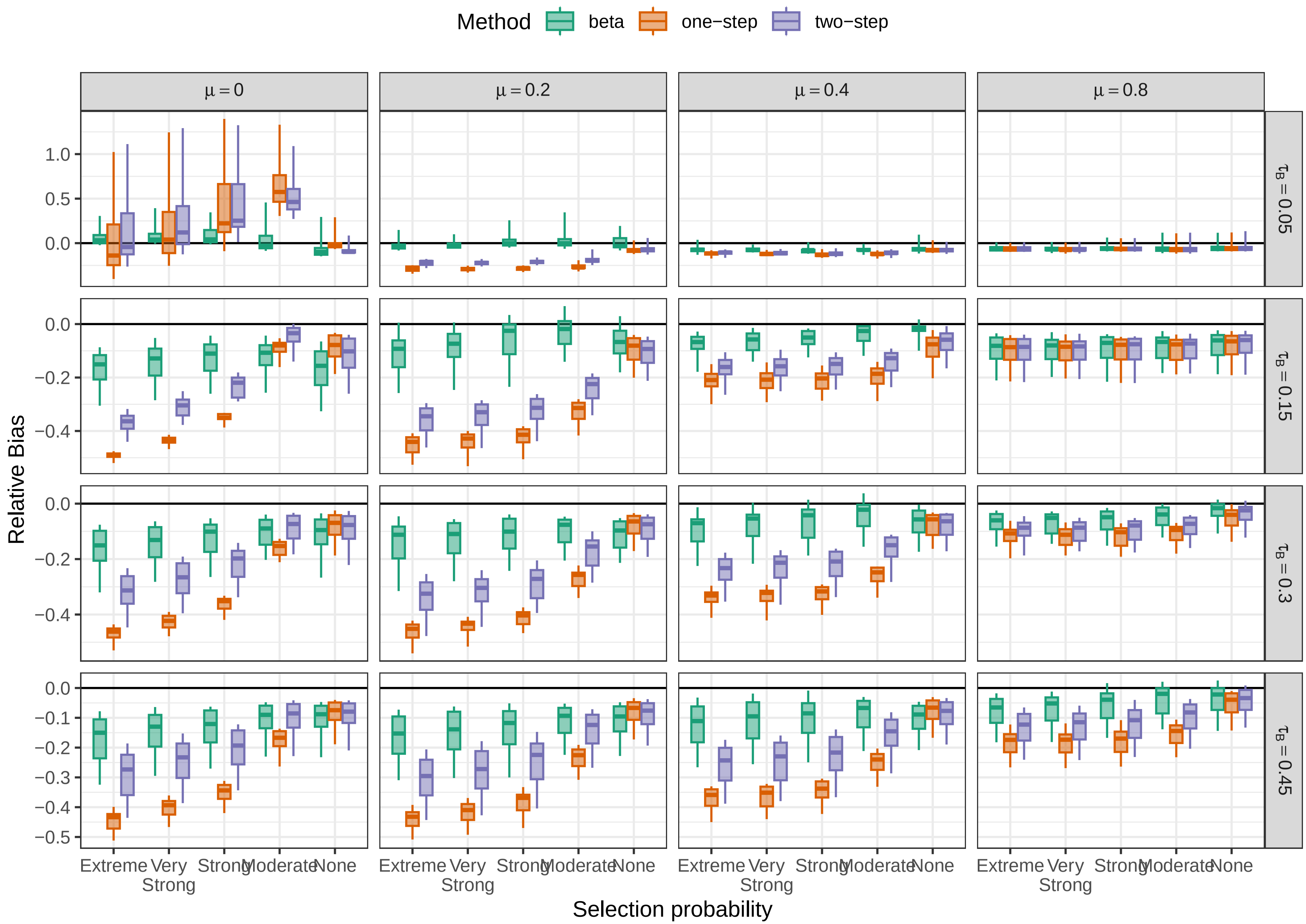

Figure C1

Relative bias of heterogeneity by method, selection probability, average SMD, and between-study heterogeneity

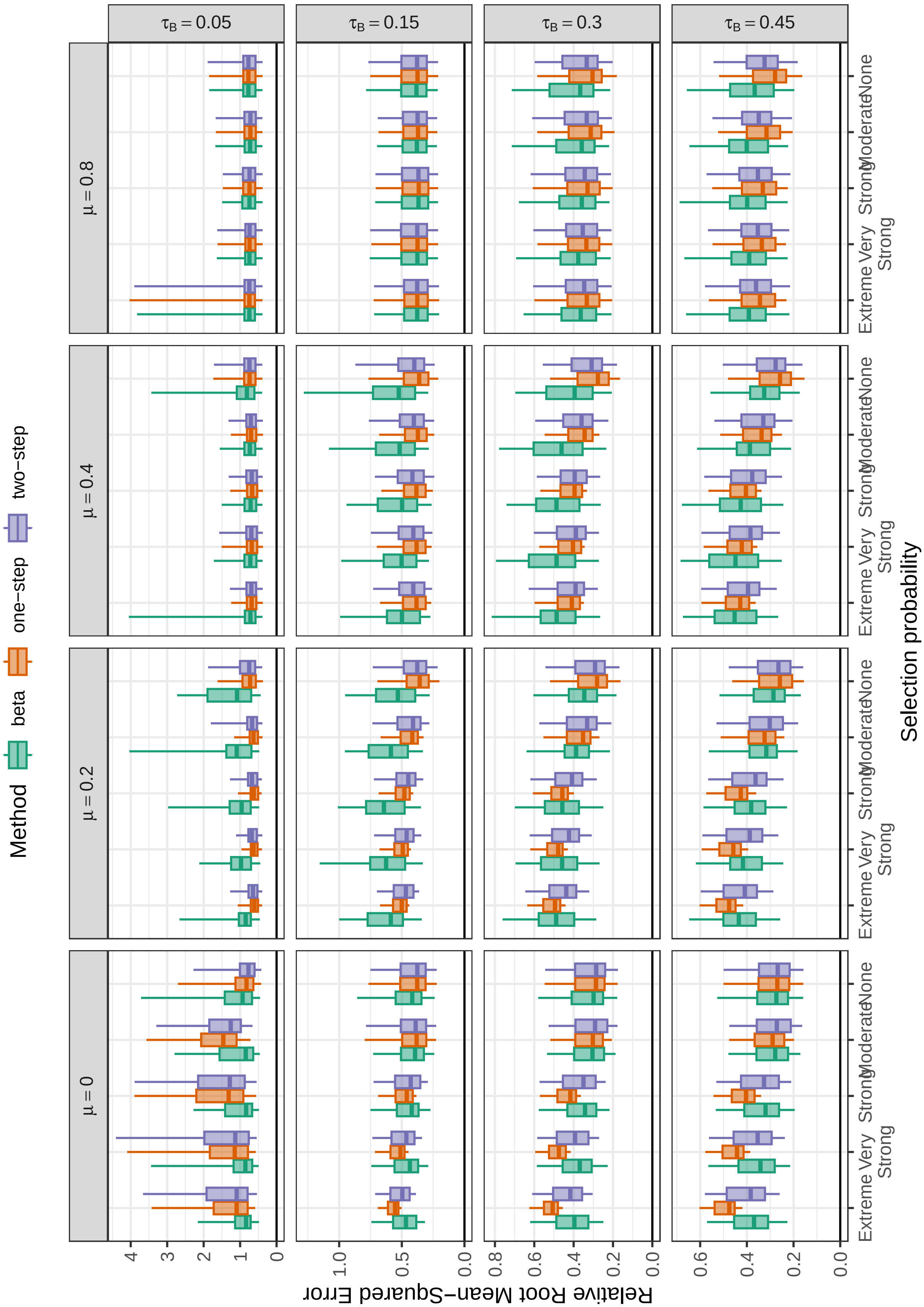

Figure C2

Relative root mean-squared error of heterogeneity by method, selection probability, average SMD, and between-study heterogeneity

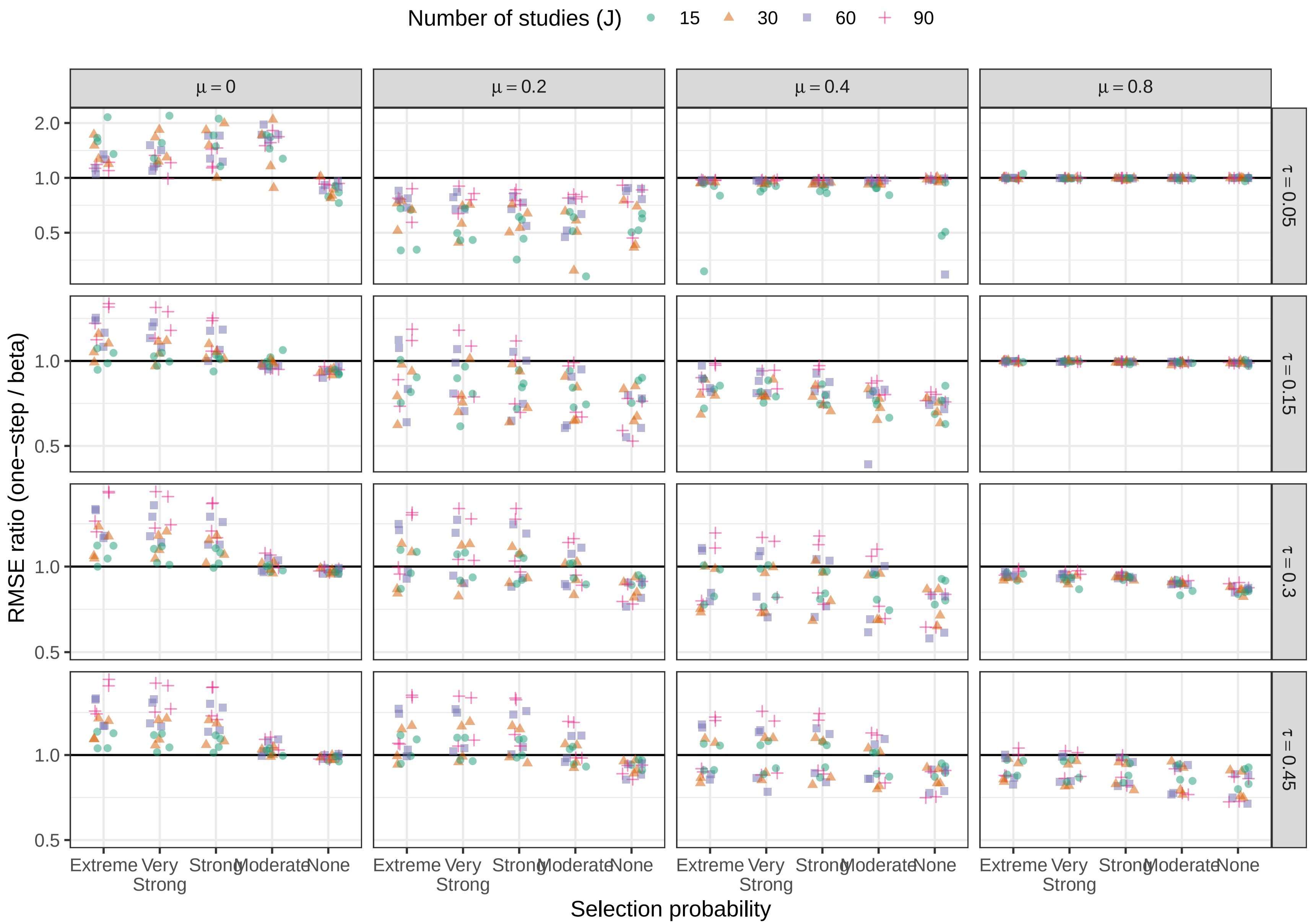

Figure C3

Ratio of root mean-squared error of heterogeneity for one-step to root mean-squared error for beta-density by selection probability, number of studies, average SMD, and between-study heterogeneity

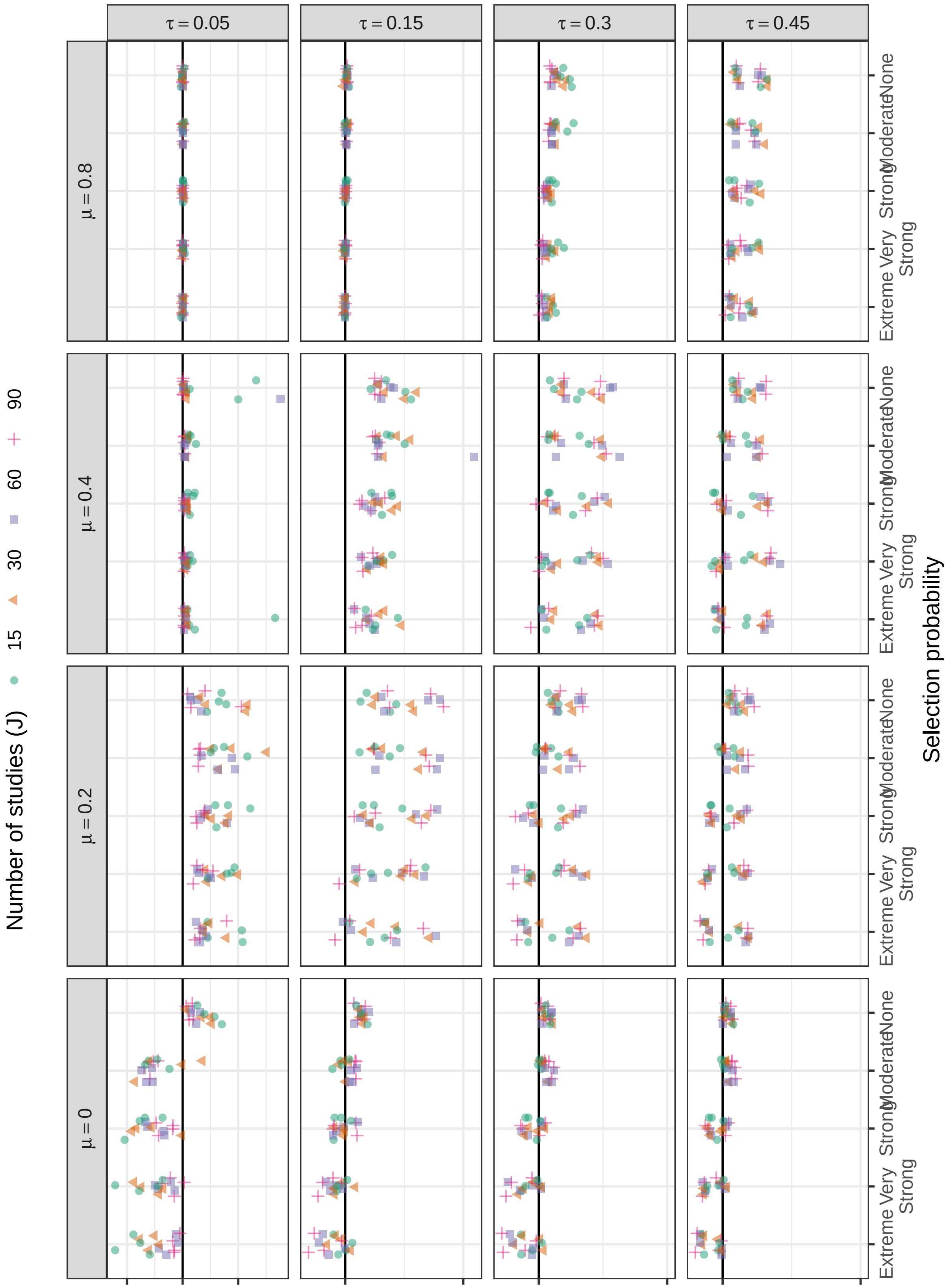

Figure C4

Ratio of root mean-squared error of heterogeneity for two-step to root mean-squared error for beta-density by selection probability, number of studies, average SMD, and between-study heterogeneity

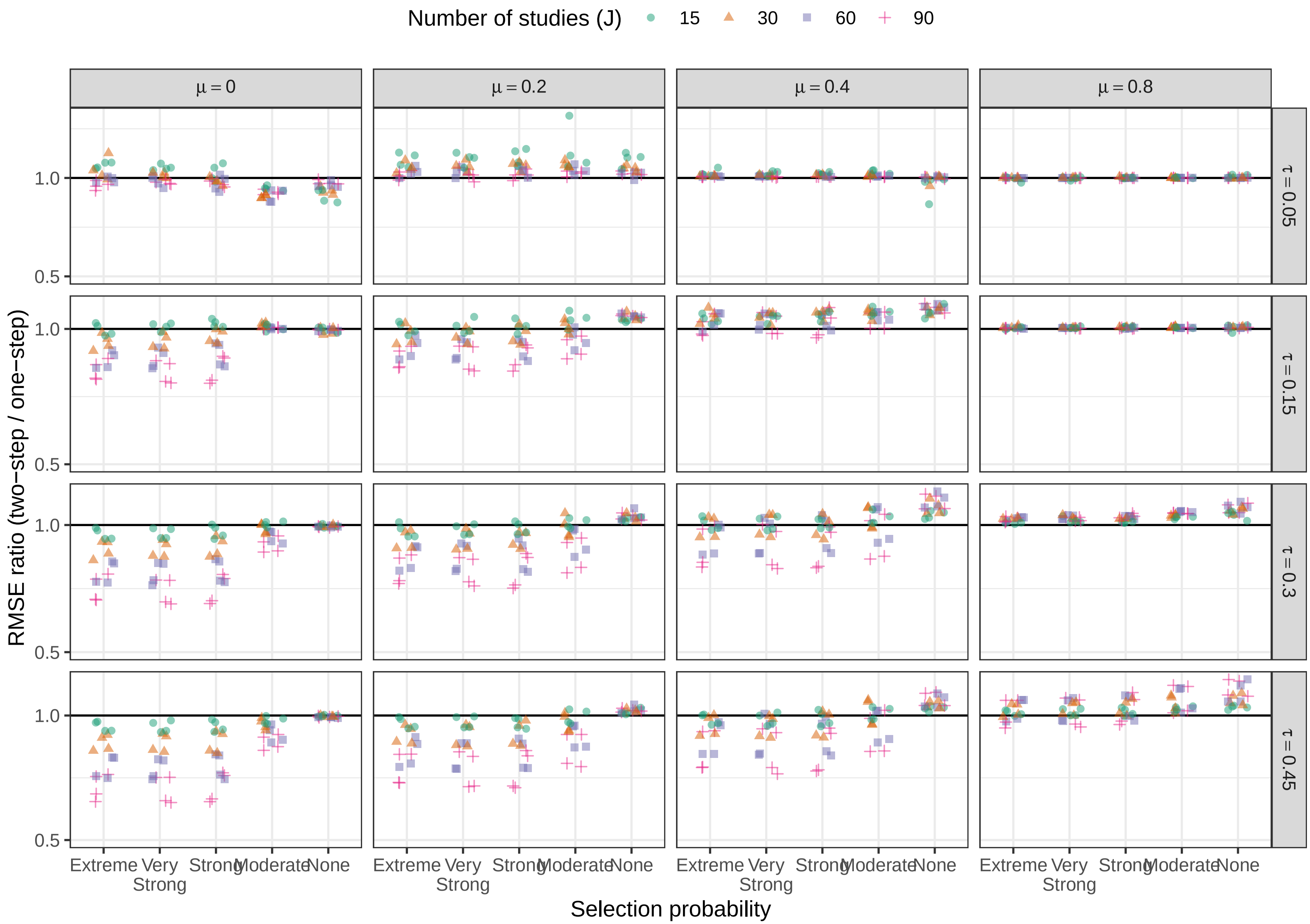

Figure C5

Ratio of root mean-squared error of heterogeneity for two-step to root mean-squared error for one-step by selection probability, number of studies, average SMD, and between-study heterogeneity

Appendix D

Additional simulation results for methods of estimating selection parameters (λ_1 & λ_2)

The beta-density selection model estimates two selection parameters: λ_1 and λ_2. Figures D1 and D2 present the relative bias and RMSE results for λ_1 and Figures D3 and D4 present the relative bias and RMSE results for λ_2. We do not plot the selection parameters for the step-function selection models because they are based on a different selection function and are, thus, not comparable to the beta-density selection parameters.[3]

The figures show that λ_1 is substantially biased when selective reporting is extreme, but relative bias decreases as the number of studies (J) increases. Relative bias also decreases as the severity of selective reporting decreases, reducing down to zero when selective reporting is not present. The same pattern is evident for relative RMSE. For λ_2, on the other hand, relative bias and RMSE is much closer to zero across all conditions. Relative bias and RMSE both decrease as the number of studies (J) increases. However, only relative bias decreases as the severity of selective reporting decreases; relative RMSE generally remains the same as the severity of selective reporting decreases when heterogeneity is moderate to large ($\tau \geq 0.30$) and increases somewhat when heterogeneity is low ($\tau \leq 0.15$).

[3] Step-function selection models estimate one parameter per step; thus, the one-step model estimates one selection parameter and the two-step model estimates two selection parameters.

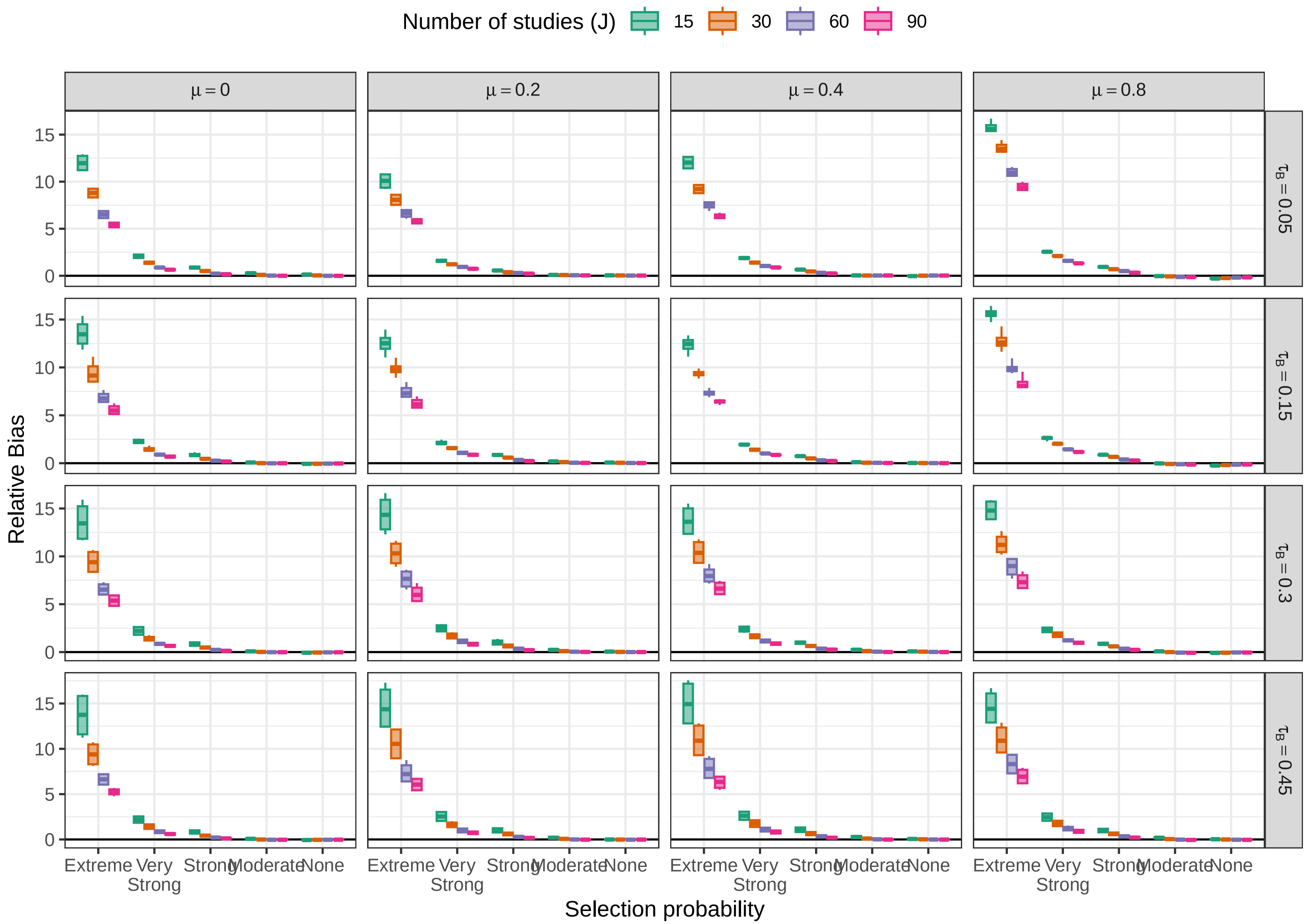

Figure D1

Relative bias of the beta-density lambda1 selection parameter by number of studies, selection probability, average SMD, and between-study heterogeneity

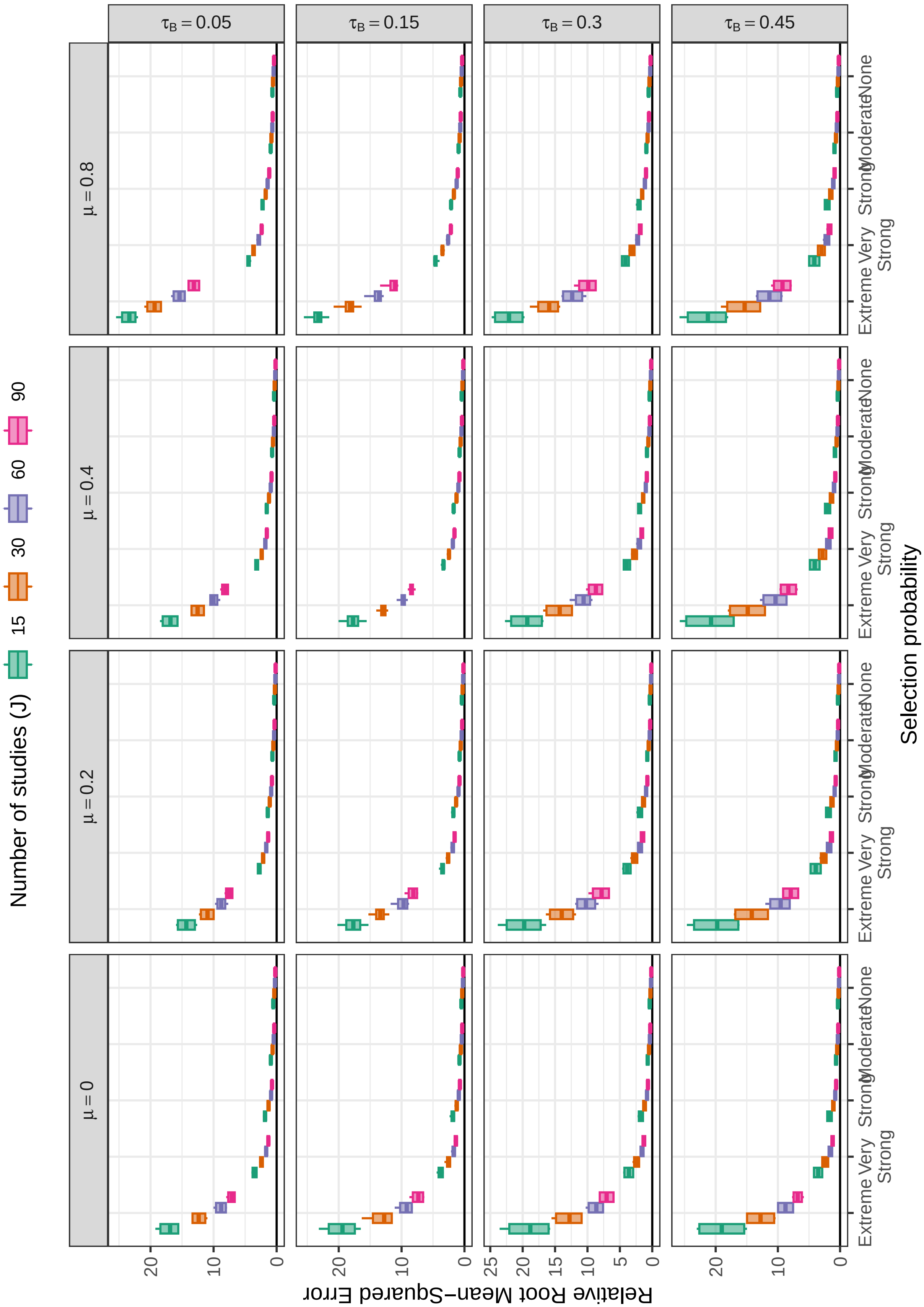

Figure D2

Relative root mean-squared error of the beta-density lambda1 selection parameter by number of studies, selection probability, average SMD, and between-study heterogeneity

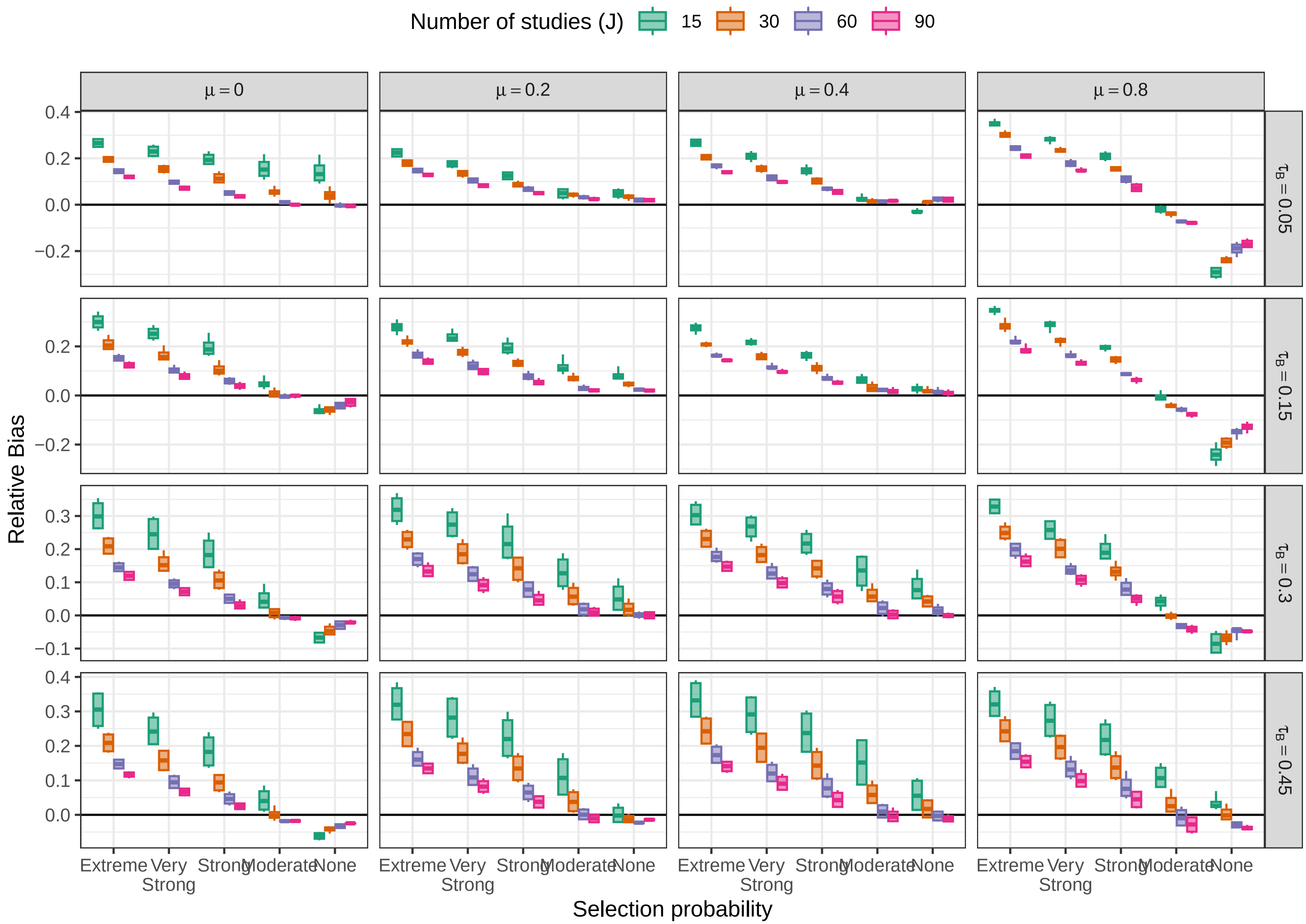

Figure D3

Relative bias of the beta-density lambda2 selection parameter by number of studies, selection probability, average SMD, and between-study heterogeneity

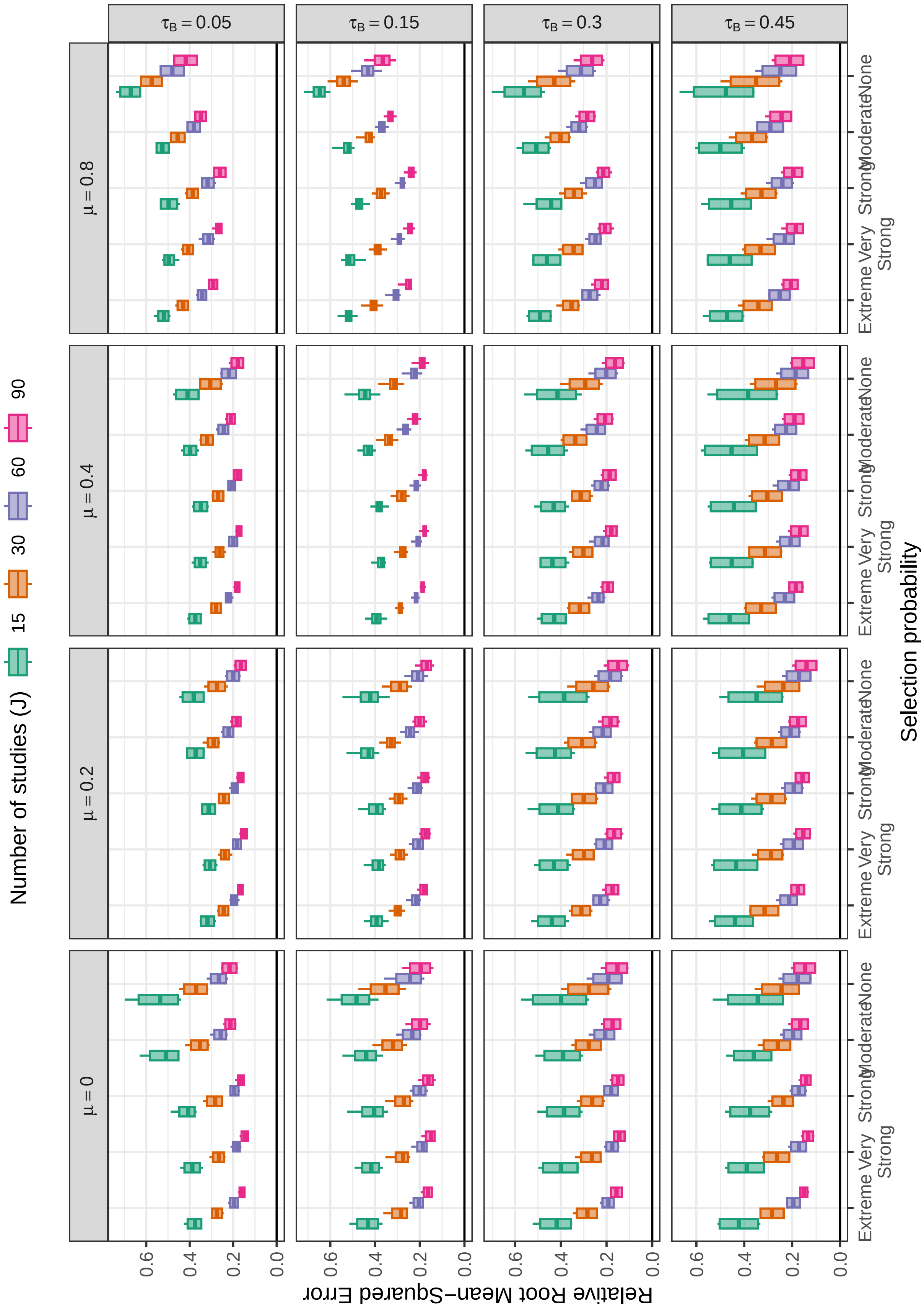

Figure D4

Relative root mean-squared error of the beta-density selection parameter by number of studies, selection probability, average SMD, and between-study heterogeneity